

Name: _____

Double Award Applied Science – Unit 1 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Marks

Year	G	F	E	D	C	Max UMS
2017	11	16	21	26	32	38
2018	8	12	16	20	25	30
2019	10	14	18	23	28	33
2022	10	14	18	23	28	34
2023	17	22	27	32	38	44

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – NEW

3445U10-1



APPLIED SCIENCE (Double Award)
Unit 1: Energy, Resources and the Environment
FOUNDATION TIER

WEDNESDAY, 14 JUNE 2017 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	3	
2.	8	
3.	13	
4.	14	
5.	18	
6.	19	
Total	75	

3445U101
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

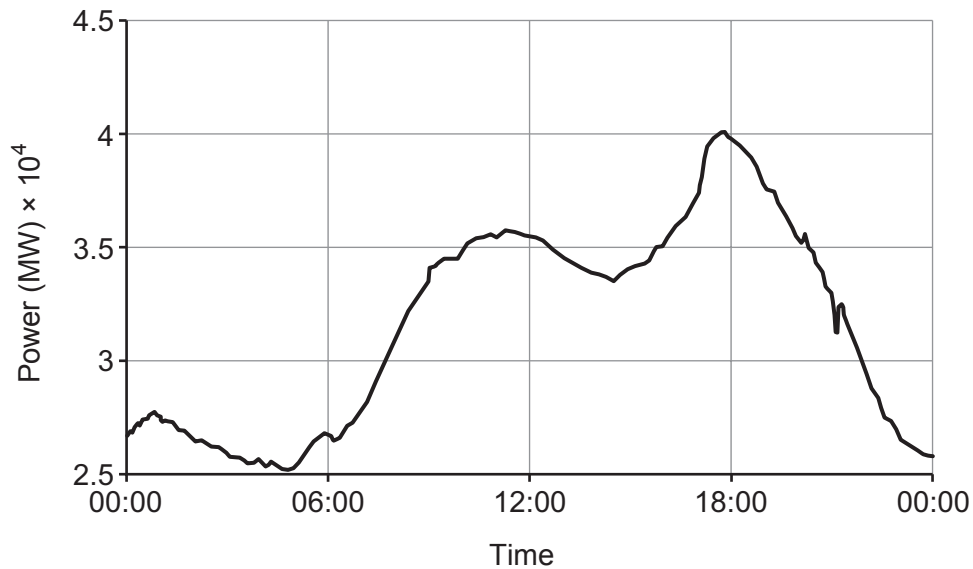
Question 5(a)(ii) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 20.

Answer **all** the questions in the spaces provided.

1. The graph below shows how demand for electricity from the National Grid changed over a 24 h period on 23rd October 2011.



- (a) Describe how demand changed between 12:00 and 18:00.

[2]

.....

.....

- (b) Hydroelectric power stations only produce electricity when there is a surge in demand. However water needs to be pumped back up at some time during the 24 h period when demand is below 3×10^4 MW and falling. Mark an **X** on the graph to show where this could be done.

[1]

2. Magnesium oxide helps prevent scale formation in boilers, heat exchangers, and piping. Burning magnesium in a gas jar of oxygen produces magnesium oxide on a small scale.



- (a) The symbol for oxygen is $^{16}_8\text{O}$.

Complete the following statements about an atom of oxygen.

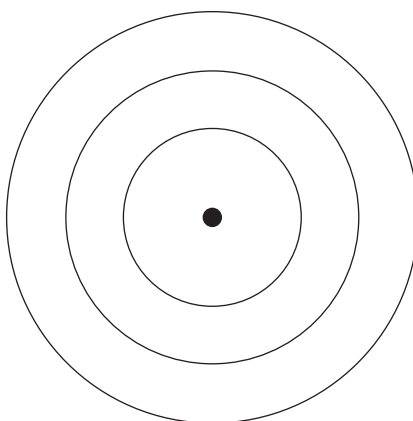
[3]

An oxygen atom contains electrons.

An oxygen atom contains neutrons.

An oxygen atom contains protons.

- (b) An atom of magnesium contains 12 electrons. Complete the diagram to show the electronic structure of a magnesium atom. [1]



- (c) State what happens to a magnesium atom when it becomes a magnesium ion.

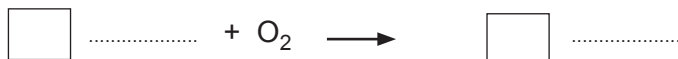
[1]

Examiner
only

- (d) **Complete** the balanced symbol equation for the following reaction.

[3]

magnesium + oxygen $\longrightarrow$ magnesium oxide



3. The halogens are a group of elements found in group 7 of the periodic table. The table below gives some information about the elements in this group.

Halogen	Atomic number	Melting point (°C)	Boiling point (°C)	Radius of the atom (units)
fluorine	9	-219	-188	(0.64) / 1.1 / 1.7
chlorine	17	(-5 / -101 / -220)	-34	0.99
bromine	35	-7	59	(1.14 / 1.45 / 1.56)
iodine	(22 / 31 / 53)	114	184	1.33
astatine	85	302	380	1.40

- (a) **Complete** the table above by **circling** the correct value in each of the brackets. One has been done as an example. [3]

- (b) Use the information in the table and your own knowledge to answer the following questions.

- (i) The boiling points of the alkali metals (group 1) decreases from the top to the bottom of their group. Describe whether the halogens follow the same trend. [2]

.....

.....

.....

- (ii) State the **two** halogens in the table which are gases at 20°C. [2]

..... and

- (iii) State which halogen is a liquid at 20°C. [1]

.....

- (iv) State what happens to the size of the halogen atoms as you go down from the top to the bottom of the group. [1]

.....

(v) **Tick (✓)** the box next to the correct statements about halogens.

[3]

Fluorine has the highest melting point

☐

The most reactive halogen is fluorine

☐

Astatine has the most electrons

☐

Iodine is a solid at 20°C

☐

The most reactive halogen is astatine

☐

Chlorine boils at a lower temperature than fluorine

☐

(c) Sodium chloride can be made by burning sodium in chlorine gas. However it is found naturally in rock salt, which can be obtained from the Earth's crust. **Tick (✓)** the box next to the correct process for extraction of rock salt from the Earth's crust.

[1]

fracking

☐

deep shaft mining

☐

fractional distillation

☐

drilling

☐

4. Rising gas prices are affecting many households. The government can help households keep their gas bills as low as possible.

Data for a typical house are shown below. The table also shows government estimates for the current usage of each method.

Part of house	Heat loss before insulation is fitted (%)	Method of reducing heat loss	Cost of method (£)	Annual saving on gas bills (£)	% of homes already using this method
roof	25	loft insulation	450	150	91
windows & doors	10	double glazing	3 900	90	70
walls	40	cavity wall insulation	1 200	250	42
floor	15	carpets	800	40	90
draughts	10	draught excluders	60	5	85

- (a) A heating engineer estimates keeping the house with no insulation at 20°C results in gas bills of £2 000 per year.

Use the information above to answer the following questions.

- (i) Calculate the cost of the heat lost **through the roof**. [2]

cost = £

- (ii) State how the engineer's estimate will change if the home is kept at 22°C. [1]

.....

- (iii) State how the engineer's estimate will change if the price of gas drops. [1]

.....

- (iv) The engineer recommends that loft insulation should be installed first.

Calculate the payback time for loft insulation. [2]

payback time = years

- (v) Which method of reducing heat loss should the government help people install by giving them a grant? Give **one** reason for your choice. [2]

.....

.....

.....

- (b) At present, many homes with central heating lack a timer, sufficient hot water tank insulation, room thermostat and thermostatic radiator valves. Upgrading all homes so they have these four features could make savings of over 25% of the energy used for heating and hot water.

Fitting new hot water tank insulation, a room thermostat and thermostatic radiator valves reduces the annual gas bill for a house from £2 000 to £1 400.

- (i) Calculate the percentage saving made to the gas bill. [2]

% saving =

- (ii) Use the equation:

$$\text{decrease in the number of units used} = \frac{\text{savings}}{\text{cost of one unit}}$$

to calculate the decrease in the number of units used. [2]
Assume each unit costs £0.20.

decrease in the number of units used = kWh

- (c) State **two** ways in which using less gas to heat the house will benefit the environment. [2]

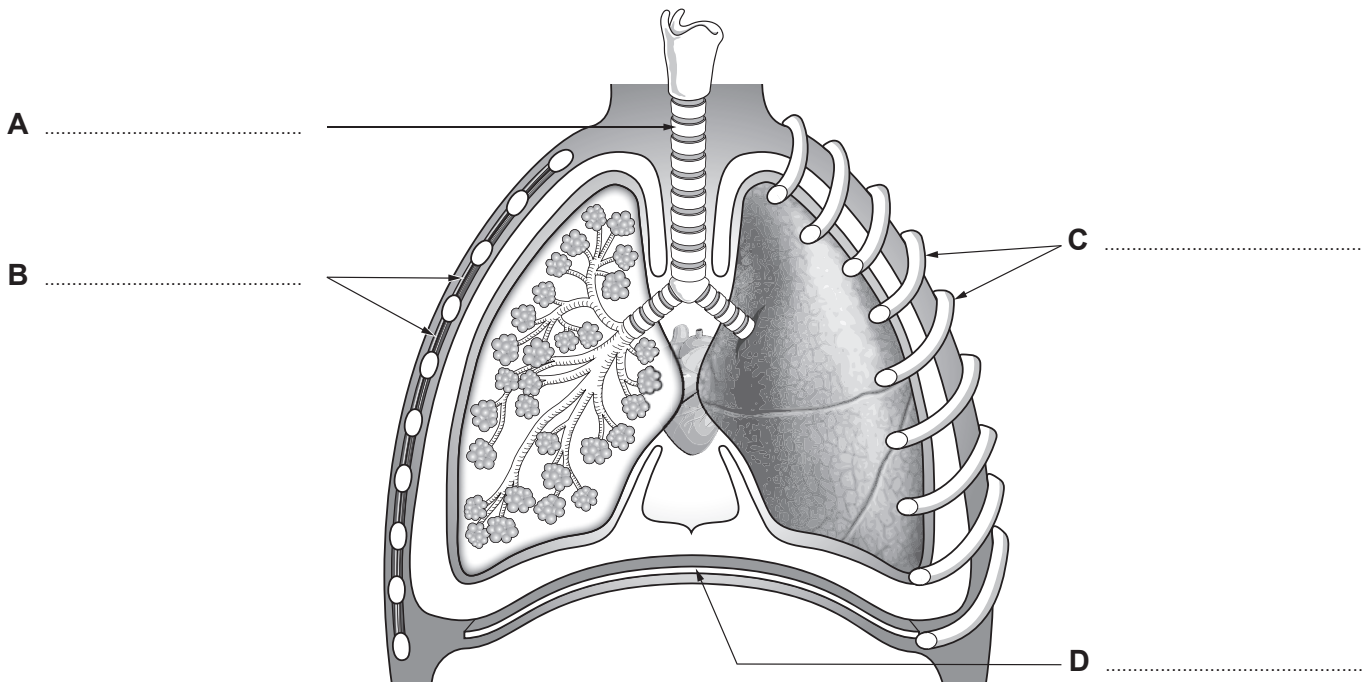
1.
2.

5. Sports science students are learning about the respiratory system.

(a) (i) Label the diagram below using words from the box.

[4]

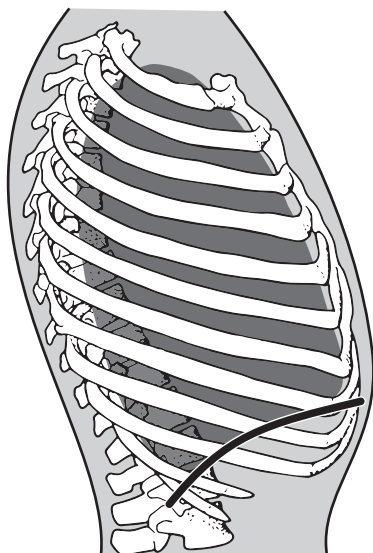
ribs intercostal muscles diaphragm lung trachea



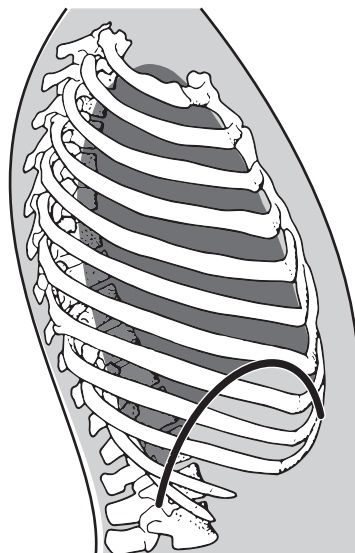
Examiner
only

- (ii) Use your knowledge and the diagrams below to describe how the changes in the diaphragm, ribcage and intercostal muscles allow you to breathe in and out.

[6 QER]



breathing in

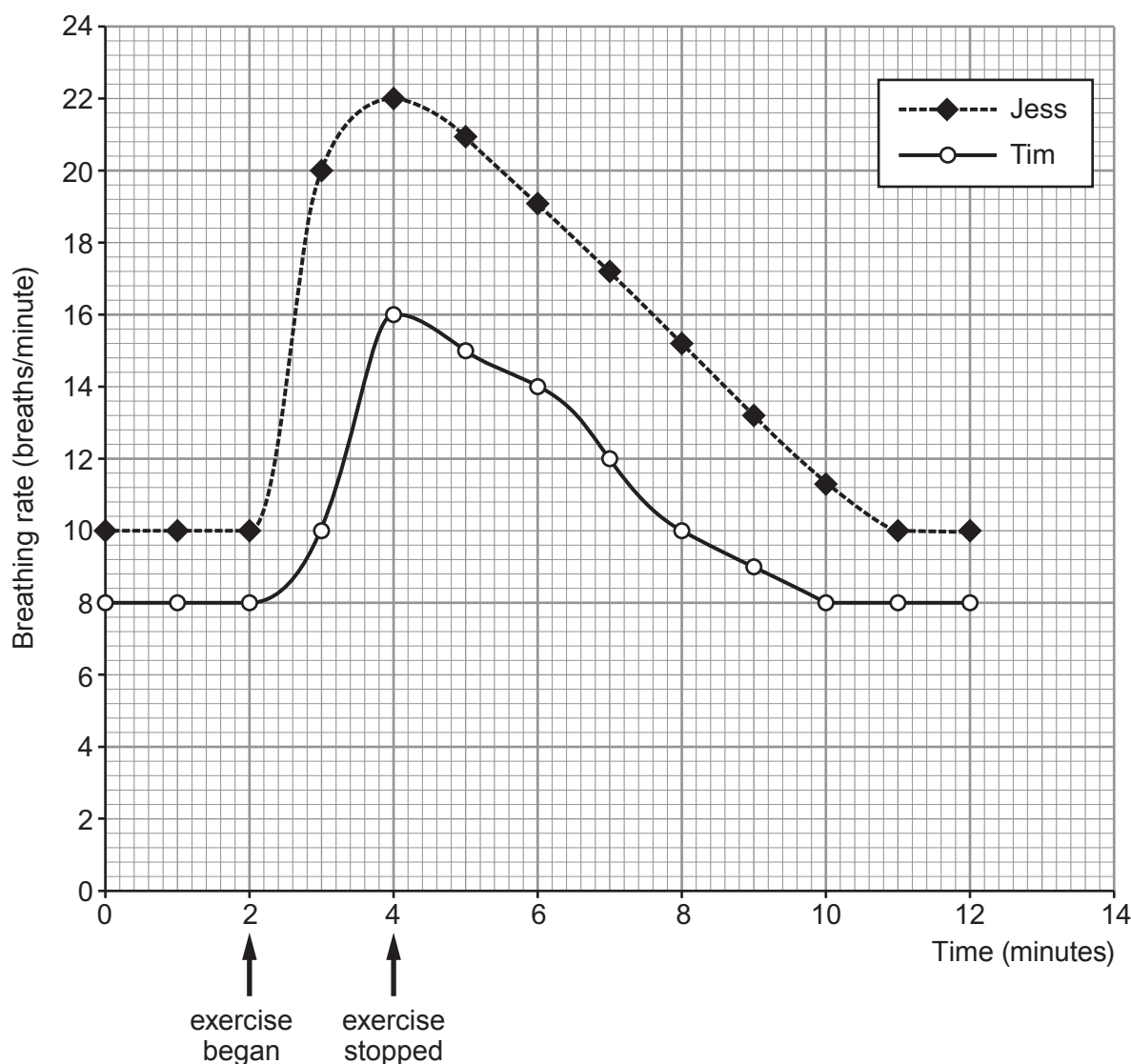


breathing out

- (b) Two students, Tim and Jess, studied the effect of exercise on breathing rate. They followed the method below:

1. record breathing rate before exercising
2. exercise for 2 minutes by running on a treadmill
3. measure the breathing rates during and after exercise up until 12 minutes

Their results are shown on the graph below.



Use the graph to complete the following sentences.

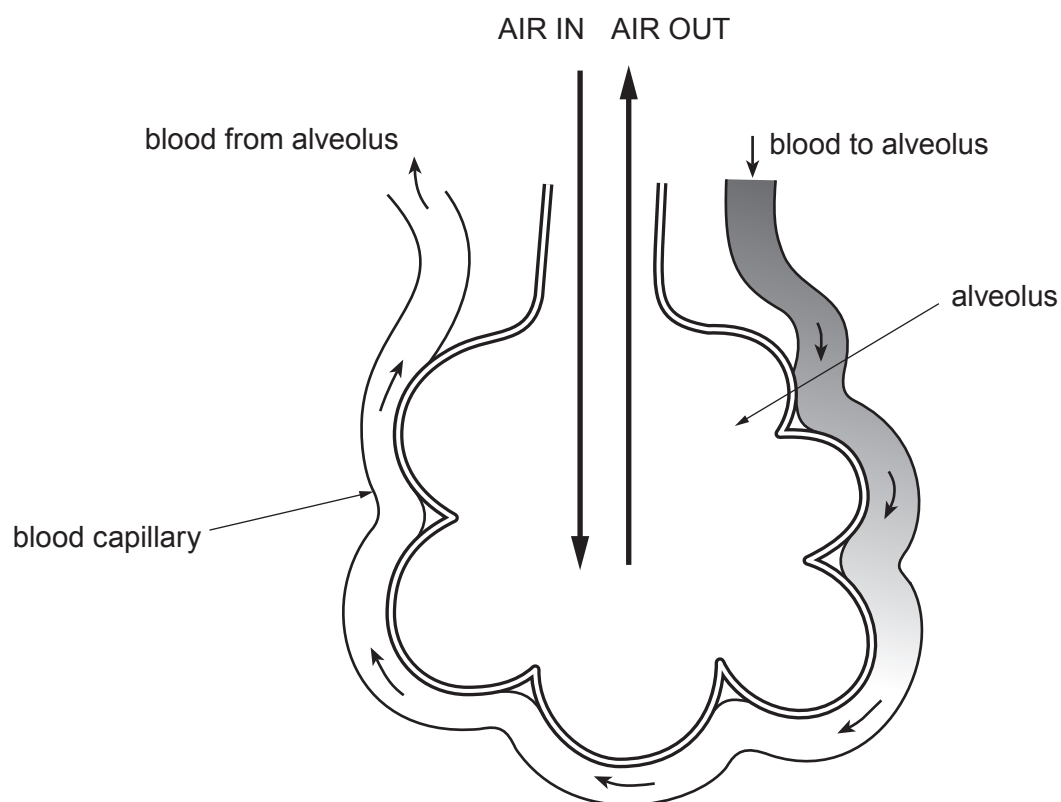
[3]

At 2 minutes, the difference between Tim and Jess' breathing rate is breaths per minute.

Between 8 and 10 minutes Tim's breathing rate drops than Jess' breathing rate.

Tim's breathing rate returns to normal at minutes.

(c) The diagram below is of an alveolus within the lung where gases are exchanged.



(i) State which gases are exchanged.

[2]

..... and

(ii) Name the process by which these gases move between the alveolus and the capillary.

[1]

.....

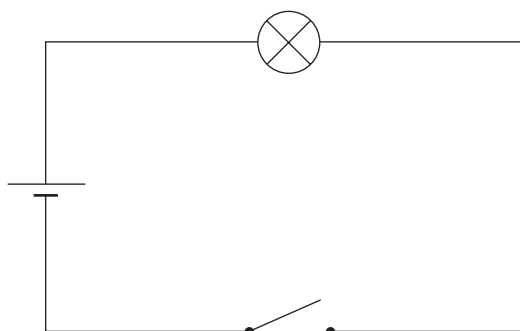
(iii) State **two** ways in which the alveoli are adapted for gas exchange.

[2]

1.

2.

6. Students are investigating how the current through a lamp varies with the voltage across it.



- (a) (i) Part of the circuit used is shown above. **Add** a variable resistor, ammeter and voltmeter to the circuit diagram. [3]
- (ii) Describe the method you would use to take a range of results. [3]

.....

.....

.....

.....

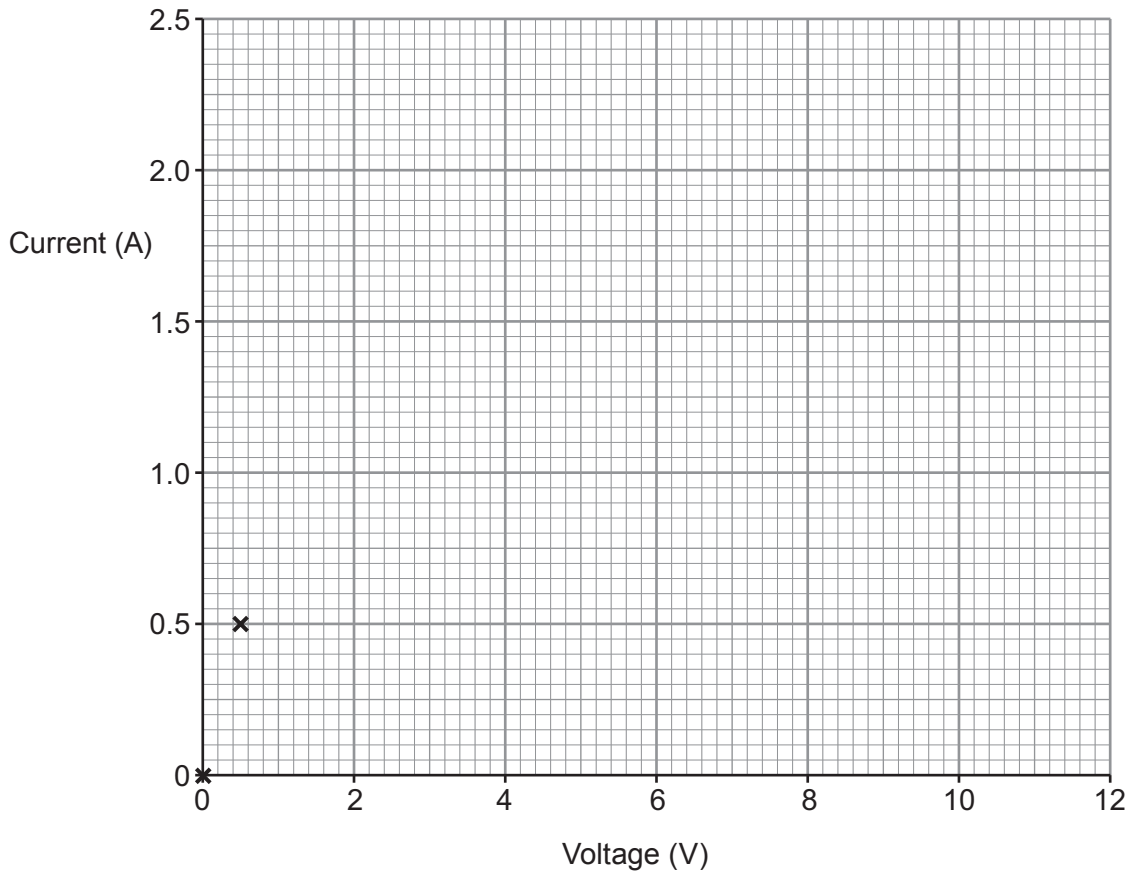
- (b) Results from the experiment are shown below.

Voltage (V)	Current (A)
0.0	0.0
0.5	0.5
2.0	0.8
4.0	1.1
6.0	1.4
8.0	1.6
10.0	1.8
12.0	1.9

- (i) Plot the data on the grid below and draw a suitable line.

[3]

The first two points have been plotted for you.



- (ii) Describe the relationship between current and voltage for the lamp.

[2]

.....

.....

.....

- (c) (i) Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the lamp at 0.5 V.

[2]

resistance = Ω

- (ii) State how the resistance of the lamp varies between 0.5 V and 12 V.

[1]

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only

- (d) (i) Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of the lamp at 0.5 V.

[2]

power = W

- (ii) State how the power of the lamp varies between 0.5 V and 12 V.

[1]

- (e) The experiment was repeated with a length of wire at constant temperature. The current through the wire was 1.5 A when the voltage across it was 12 V.

Add a line to your graph to show how current varies with voltage for this wire.

[2]

END OF PAPER

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1														4 He Helium 2																	
														19 F Fluorine 9	16 O Oxygen 8	14 N Nitrogen 7	12 C Carbon 6	11 B Boron 5													
														35.5 Cl Chlorine 17	32 S Sulfur 16	31 P Phosphorus 15	28 Si Silicon 14	27 Al Aluminium 13													
														80 Br Bromine 35	79 Se Selenium 34	75 As Arsenic 33	73 Ge Germanium 32	70 Ga Gallium 31													
														127 I Iodine 53	128 Te Tellurium 52	122 Sb Antimony 51	119 Sn Tin 50	115 In Indium 49													
														210 At Astatine 85	210 Po Polonium 84	209 Bi Bismuth 83	207 Pb Lead 82	204 Tl Thallium 81													
														222 Rn Radon 86						201 Hg Mercury 80	197 Au Gold 79	195 Pt Platinum 78	192 Ir Iridium 77	190 Os Osmium 76	186 Re Rhenium 75	184 W Tungsten 74	181 Ta Tantalum 73	179 Hf Hafnium 72	139 La Lanthanum 57	137 Ba Barium 56	133 Cs Caesium 55
																			112 Cd Cadmium 48	108 Ag Silver 47	106 Pd Palladium 46	103 Rh Rhodium 45	101 Ru Ruthenium 44	99 Tc Technetium 43	96 Mo Molybdenum 42	93 Nb Niobium 41	89 Y Yttrium 39	88 Sr Strontium 38	86 Rb Rubidium 37		
														65 Zn Zinc 30	63.5 Cu Copper 29	59 Ni Nickel 28	59 Co Cobalt 27	56 Fe Iron 26	55 Mn Manganese 25	52 Cr Chromium 24	51 V Vanadium 23	48 Ti Titanium 22	45 Sc Scandium 21	40 Ca Calcium 20	39 K Potassium 19						
																			227 Ac Actinium 89	226 Ra Radium 88	223 Fr Francium 87										

Key

1

H

Hydrogen

1

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0



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S18-3445U10-1

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

FOUNDATION TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	13	
3.	10	
4.	13	
5.	9	
6.	10	
7.	9	
Total	75	

3445U101
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **3(b)(i)** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 24.

Answer **all** questions.

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1. Raw materials are extracted from the Earth's crust or atmosphere by different methods.

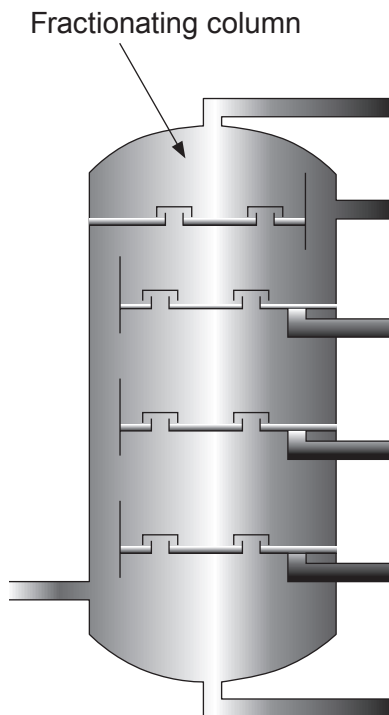
- (a) Identify the correct extraction method for each raw material by drawing a line.
One has been done for you.

[2]

Raw material	Extraction method
crude oil	fractional distillation
shale gas	drilling
haematite (a metal ore)	surface mining
oxygen	fracking

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03

- (b) A fractionating column is used to separate the hydrocarbons in crude oil. The table shows information about some of the fractions obtained.



Fraction	Carbon chain length	Boiling range (°C)
Refinery gases	1-4	<40
Petrol	5-8	40-110
Naphtha	8-10	110-180
Kerosene	10-16	
Diesel	16-20	260-320
Fuel oil	20-50	320-400
Bitumen	>50	>400

- (i) **Complete** the table. [1]
- (ii) Name the process which is used to separate crude oil into its fractions. [1]
-
- (iii) State what happens to the crude oil immediately before it enters the column. [1]
-
- (iv) State the fraction that a hydrocarbon with a boiling point of 290 °C will belong to. [1]
-

(c) Complete the statements below by underlining the correct term in the brackets.

(i) The temperature (**increases, stays the same, decreases**) going up the fractionating column. [1]

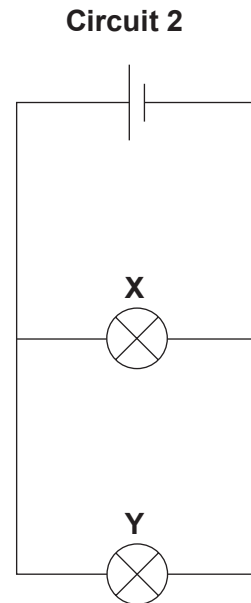
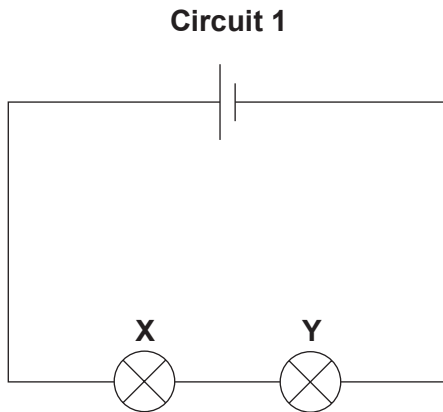
(ii) The boiling point of a hydrocarbon (**increases, decreases, stays the same**) as the chain length increases. [1]

(d) The fractionating column produces hydrocarbons that can be further processed and used to make plastics.

Complete the table to show how polyethene, polychloroethene (PVC) and polytetrafluoroethene (PTFE) are made. [3]

Monomer			Polymer	
Name	Molecular formula	Structural formula	Name	Structure
Ethene	C_2H_4	$ \begin{array}{c} H & & H \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ H & & H \end{array} $	polyethene	
Chloroethene	C_2H_3Cl		PVC	$ \left[\begin{array}{cc} Cl & H \\ & \\ -C & -C- \\ & \\ H & H \end{array} \right]_n $
Tetrafluoroethene		$ \begin{array}{c} F & & F \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ F & & F \end{array} $	PTFE	$ \left[\begin{array}{c} F \\ \\ -C- \\ \\ F \end{array} \right]_n $

2. A student is studying the properties of different electrical circuits. **X** and **Y** are identical lamps. The student built the following circuits.



- (a) Complete the sentences using the terms in the box.
The terms can be used once, more than once or not at all.

series	less than	parallel	the same as
greater than	simple	current	voltage

- (i) The lamps in **circuit 1** are connected in
 The current through lamp **X** isthe current through lamp **Y**.
 [2]
- (ii) The lamps in **circuit 2** are connected in
 The current through lamp **X** isthe current through lamp **Y**.
 [2]

(b) In **circuit 1** the voltage supplied is 9 V. Each lamp has a resistance of 3 Ω.

(i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in this circuit.

[3]

Current = A

(ii) State what would happen to the current in **circuit 1** if lamp **Y** is removed and the circuit was built with only lamp **X**. [1]

.....

(c) In **circuit 2** the voltage supplied is also 9 V and each bulb has a current of 3 A.

Use the equation

$$\text{power} = \text{voltage} \times \text{current}$$

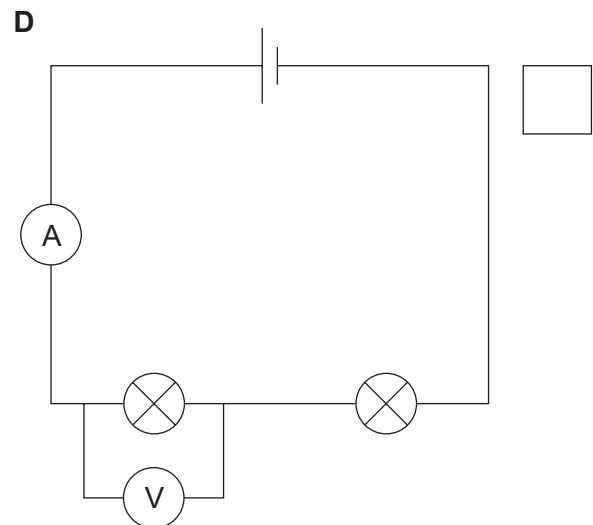
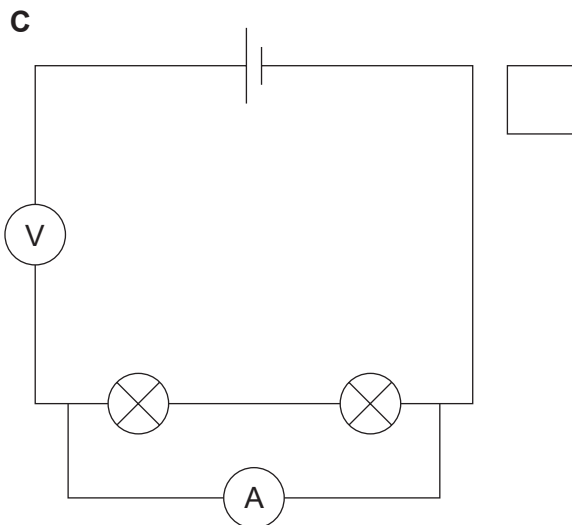
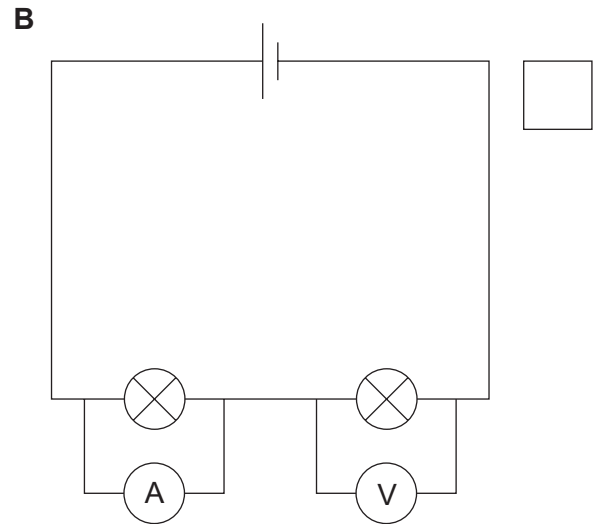
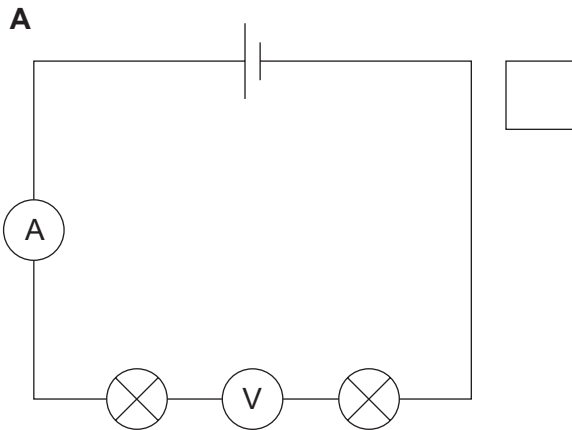
to calculate the power of lamp **X**.

[2]

Power = W

- (d) The student set up a circuit to measure the current and voltage correctly.

Use a tick (✓) to identify the circuit diagram that should have been used to measure the voltage. [1]



- (e) State **two** reasons why lamps are connected in parallel rather than in series in the home. [2]

1.
2.

3. Water treatment in the UK is an important process in making water safe to drink.

- (a) In an experiment to determine the hardness of water in different parts of the country, different technicians tested water samples from Bristol, Birmingham, Manchester and Southampton.

They divided the water samples from each city into two and carried out Test A and Test B on each portion.

Test A

For the **first** portion they followed the method below.

1. Measure 15 cm^3 of water sample into a boiling tube.
2. Add 1 cm^3 of soap solution, insert a stopper then shake vigorously for 5 seconds.
3. Repeat step 2 until lather forms that lasts for 30 seconds. Record the total volume of soap solution added.
4. Repeat steps 1-3 with 15 cm^3 of the other samples of water.

Test B

For the **second** portion they followed the method below.

- Boil the water for 10 minutes.
- Then follow steps 1-4 in **Test A**.

- (i) State the dependent variable in the experiment.

[1]

.....

- (ii) How could the reproducibility of the experiment be checked?

[1]

.....

3445U101

- (b) The results of the experiment are shown below.

Sample	Volume of soap used to form lather (cm ³)	
	Test A (without boiling)	Test B (after boiling)
Birmingham	1	1
Southampton	11	1
Bristol	12	12
Manchester	1	1

- (i) Explain how the technicians used the results to make conclusions about the type of water in **each city**. [6 QER]

- (ii) Hard water can cause problems in the home.

State **two** problems hard water can cause.

[2]

1.
2.

4. Enzymes are involved in all metabolic reactions.

(a) (i) State the **type** of biological molecule that all enzymes are made of.

[1]

.....

Enzymes can break down larger molecules into smaller units in the digestive system.

mouth	amylase	lipase	liver
large intestine	amino acids	fibre	stomach

Enzyme	Larger molecules	Smaller units	Site of action
Carbohydrase	Starch	Glucose	 and small intestine
.....	Fats and oils	Fatty acids and glycerol	small intestine
Protease	Protein		 and small intestine

(ii) **Complete the table** using only the words in the box.

[4]

- (b) Students investigated the effect of temperature on the digestion of starch.

The students mixed starch and amylase (a carbohydrase) at different temperatures.

They used iodine to test for the presence of starch. Iodine changes colour from brown to blue-black if starch is present.

The results of the students' investigation are shown below.

Temperature of the solution (°C)	Colour of the solution					
	at start	after 1 min	after 2 min	after 3 min	after 4 min	after 5 min
20	blue-black	blue-black	blue-black	blue-black	blue-black	blue-black
30	blue-black	blue-black	blue-black	blue-black	blue-black	brown
40	blue-black	brown	brown	brown	brown	brown
50	blue-black	blue-black	brown	brown	brown	brown
60	blue-black	blue-black	blue-black	brown	brown	brown
70	blue-black	blue-black	blue-black	blue-black	brown	brown
80	blue-black	blue-black	blue-black	blue-black	blue-black	blue-black

- (i) Describe how temperature affects the digestion of starch between **20 °C** and **70 °C**. [3]

.....

.....

.....

.....

.....

.....

- (ii) Explain the results at **80 °C**. [2]

.....

.....

.....

(c) When starch is completely broken down by amylase the final product is glucose.

(i) Describe the test that the students need to perform to show the presence of glucose. [2]

.....

.....

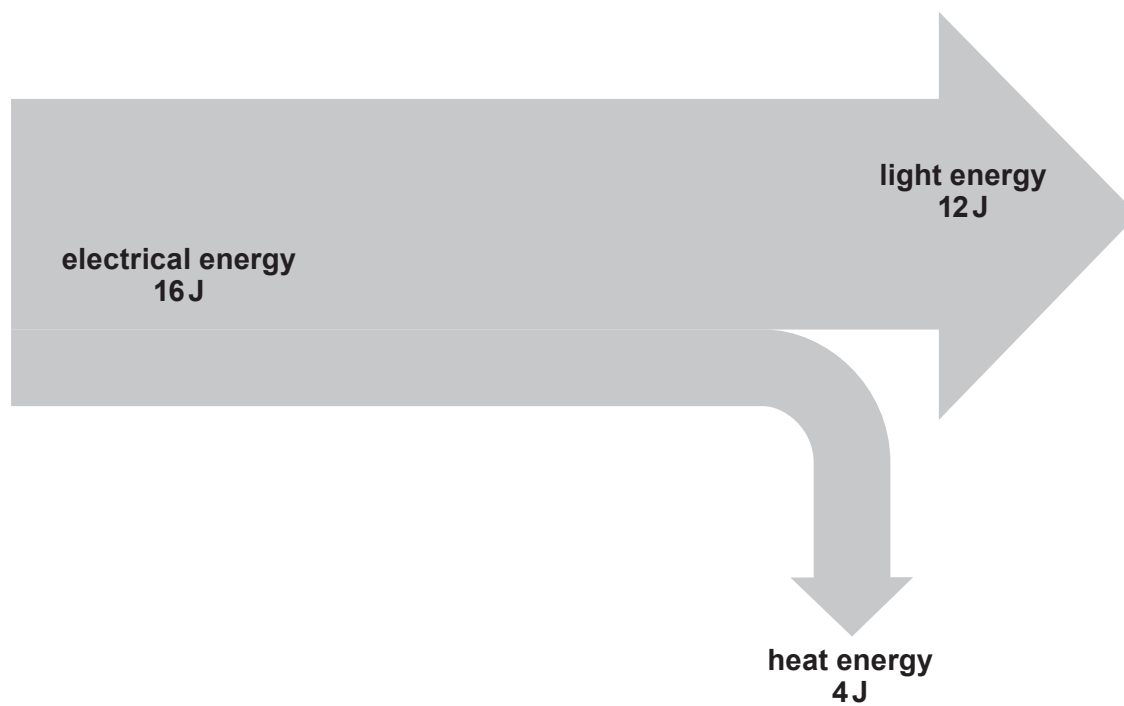
.....

(ii) State the colour **change** that shows that glucose is present. [1]

.....

5. Lighting a home accounts for 18 % of a typical house's electricity bill.

(a) The diagram below shows energy flow through a LED lamp.



Use the equation:

$$\% \text{ Efficiency} = \frac{\text{energy usefully transferred}}{\text{total energy}} \times 100$$

to calculate the efficiency of the LED lamp.

[2]

% Efficiency =

- (b) The Energy Trust is an organisation in the UK that encourages householders to become more efficient in their energy use and to help to reduce their energy costs. The table below compares traditional filament and LED lamps.

	LED	Traditional filament lamp
cost of 1 lamp	£2.50	£1
lamp lifespan (hours)	50 000	1000
number of lamps used in 50 000 hours	1	
cost of lamps used in 50 000 hours	£2.50	£
power per lamp (kW)	0.01	0.06
units used in 50 000 hours (units = power $\times$ 50 000)		3 000
cost of one unit	20p	20p
cost of electricity for light for 50 000 hours = units used $\times$ cost of unit	£	£600
total cost of light for 50 000 hours	£102.50	£

- (i) **Complete** the table.
(space for working)

[5]

- (ii) The homeowner refuses to buy LED lamps because they are more expensive and he thinks he will lose money compared to using traditional filament lamps.

Explain whether you agree with the homeowner.

[2]

.....

.....

.....

6. Students are making the fertiliser potassium nitrate in the laboratory. Most fertilisers are formed by the reaction of an acid and an alkali.

(a) Complete the word equation for the formation of potassium nitrate. [2]

..... + potassium hydroxide $\longrightarrow$ potassium nitrate +

- (b) To make the fertiliser, it is important that the students are able to exactly neutralise the acid with alkali. To do this the students carefully measured 25 cm^3 of acid into a flask and titrated just enough alkali to neutralise the acid.

The results of the experiment are shown below.

Trial	Volume of alkali added at endpoint (titre) (cm^3)
1	20.5
2	20.8
3	18.5
4	20.0
5	19.9

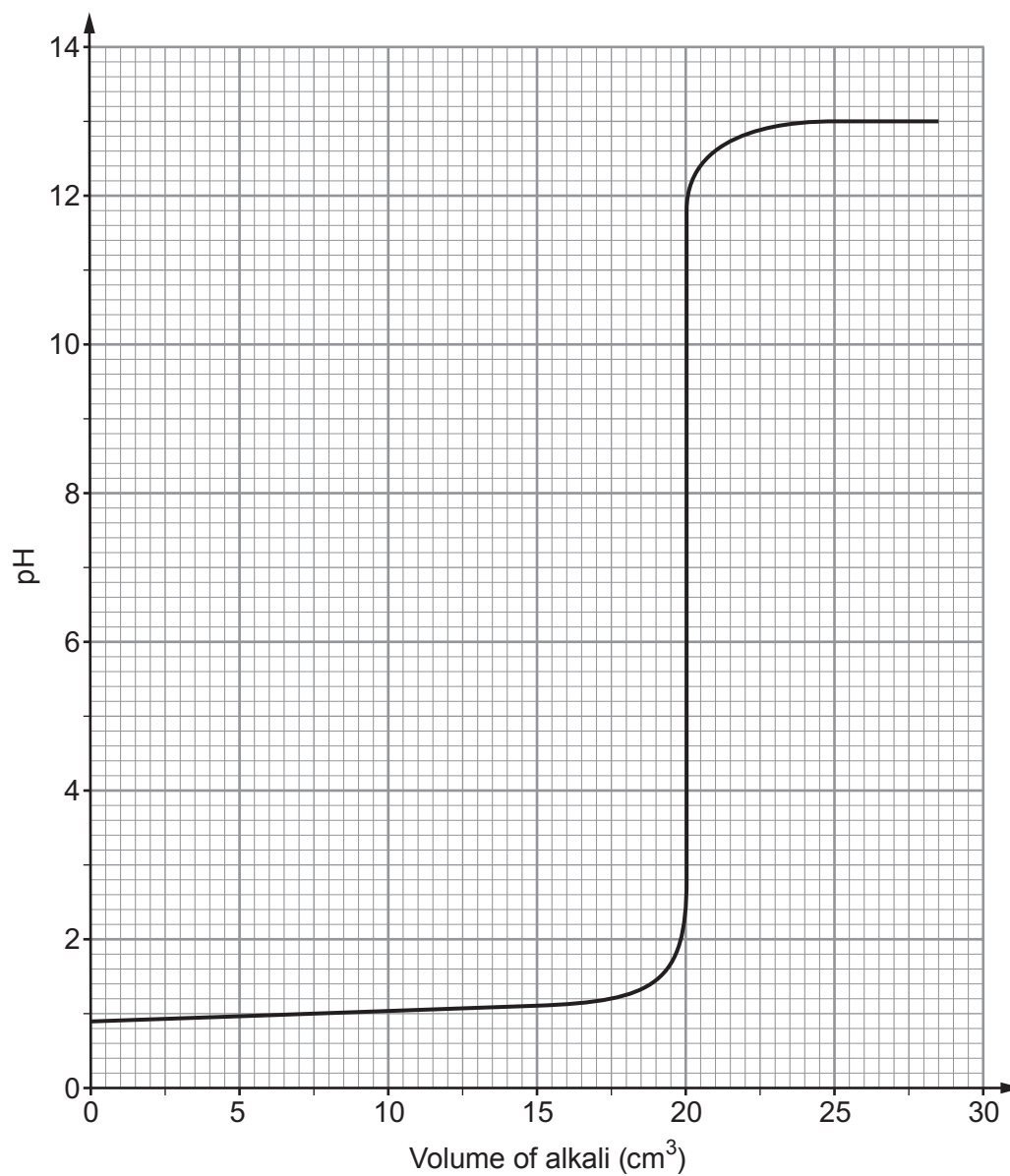
- (i) The reading for trial 3 was considered to be an anomalous result. Give a reason why. [1]

.....

- (ii) Calculate the mean titre. [1]

Mean titre = cm^3

- (c) The graph below shows the pH changes, measured by a pH probe, that occurred during trial 4 of the experiment.



Explain the pH changes shown on the graph.

[3]

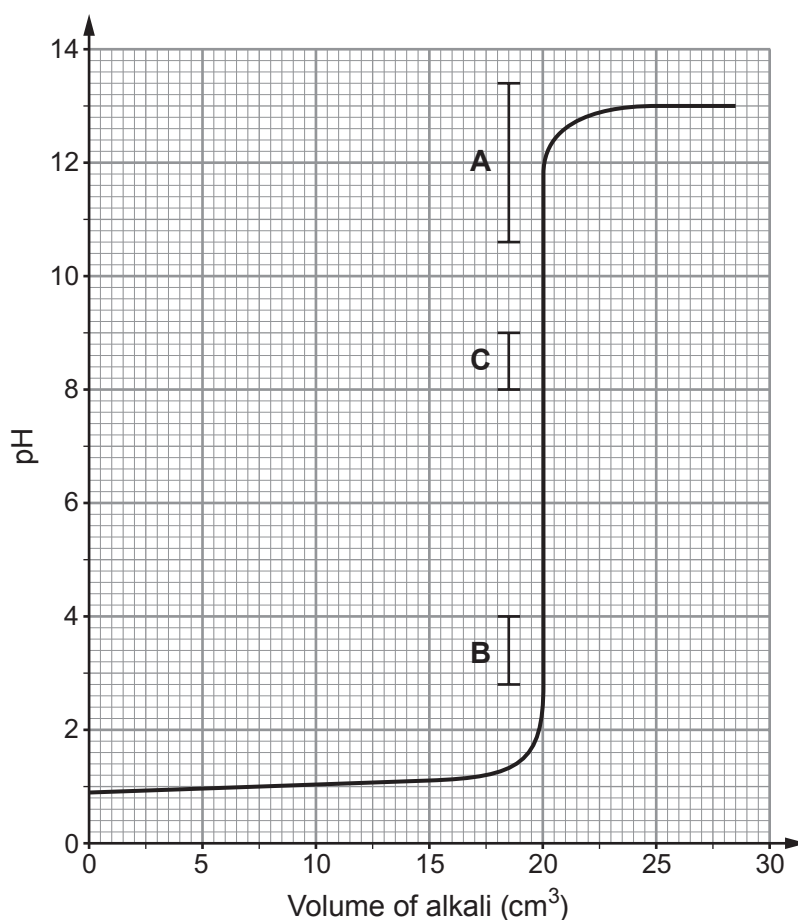
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- (d) Not all the students had a pH probe. Some had to use an indicator. The pH range of three different indicators is marked on the graph as **A**, **B**, and **C**.



- (i) Use the information in the table below to match the range to each indicator. [1]

Indicator	pH range	Colour change	Range on the graph (A , B , or C ?)
methyl blue	10.6 - 13.4	blue to violet	
phenolphthalein	8.0 - 9.0	colourless to pink	
dinitrophenol	2.8 - 4.0	colourless to yellow	

- (ii) Explain which indicator is not suitable for this experiment. [2]

.....

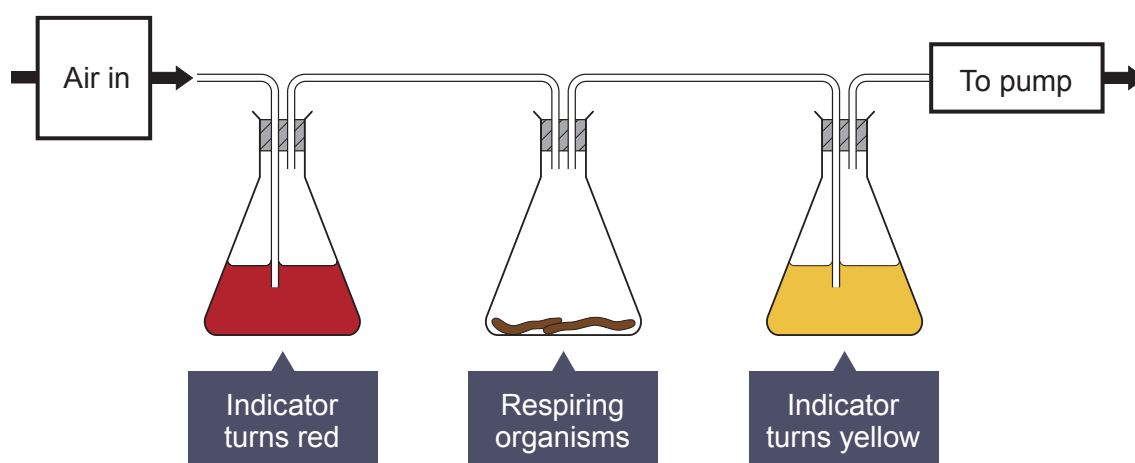
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7. Hydrogen carbonate indicator can be used to measure the rate of respiration.

Concentration of carbon dioxide (%)	Colour of hydrogen carbonate indicator
less than 0.04	purple
0.04 (normal concentration in air)	red
greater than 0.04	yellow

The apparatus below can be used to compare the rate of respiration in small animals.



A group of students used different respiring organisms and recorded the time taken for the hydrogen carbonate indicator to turn yellow. The apparatus was kept at a constant temperature of 20 °C.

The results are shown in the table below.

Organism	Time taken for the hydrogen carbonate indicator to turn yellow (min)
earthworm	22
woodlouse	18
grasshopper	10
mouse	3

- (a) Explain why the hydrogen carbonate indicator changed colour in this experiment. [2]

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- (b) Give **two** reasons why the hydrogen carbonate indicator turned yellow fastest with the mouse. [2]

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- (c) The experiment was performed at a temperature of 20 °C. [2]

State **two** other variables that need to be controlled when comparing the respiration rates of different organisms.

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- (d) Describe how you would set up a control for this experiment and state the expected result. [3]

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END OF PAPER

	Group	3	4	5	6	7	0
1	2						
1	2						

Key

A_r	Symbol	Name	Z
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Surname	Centre Number	Candidate Number
Other Names		0



GCSE

3445U10-1



APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

FOUNDATION TIER

WEDNESDAY, 12 JUNE 2019 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	13	
2.	8	
3.	16	
4.	7	
5.	12	
6.	11	
7.	8	
Total	75	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the continuation pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

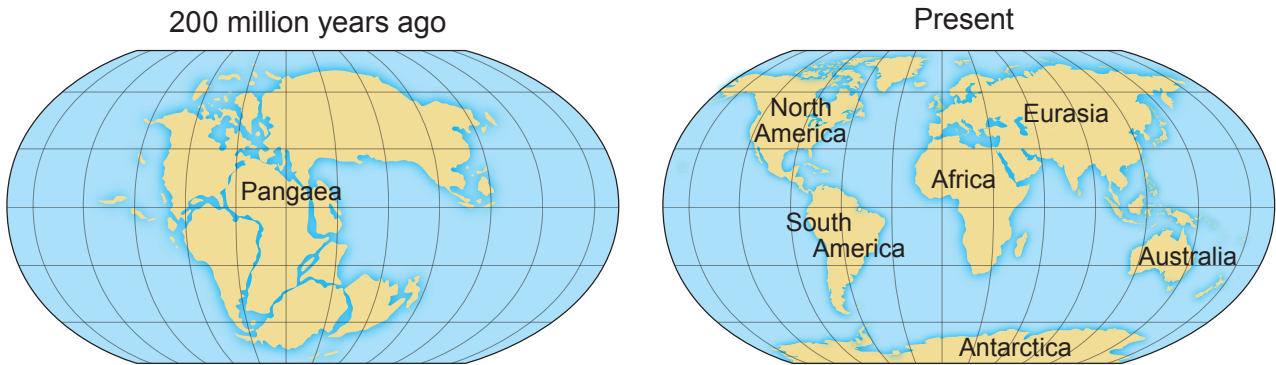
Question **3(b)** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 24.

Answer all questions.

1. In 1912 Alfred Wegener suggested that all of the continents on Earth were once joined together in one supercontinent called Pangaea. These continents have since drifted apart. Wegener suggested a hypothesis called 'Continental Drift'.



- (a) Tick (✓) three pieces of evidence that support Wegener's theory.

[3]

Mountain ranges always form at the edge of each continent.

☐

The edges of continents fit like jigsaw pieces.

☐

The oceans get deeper at the continental edges.

☐

Similar rocks of the same age are found on facing sides of different continents.

☐

Similar shaped rivers are seen at each continental shoreline.

☐

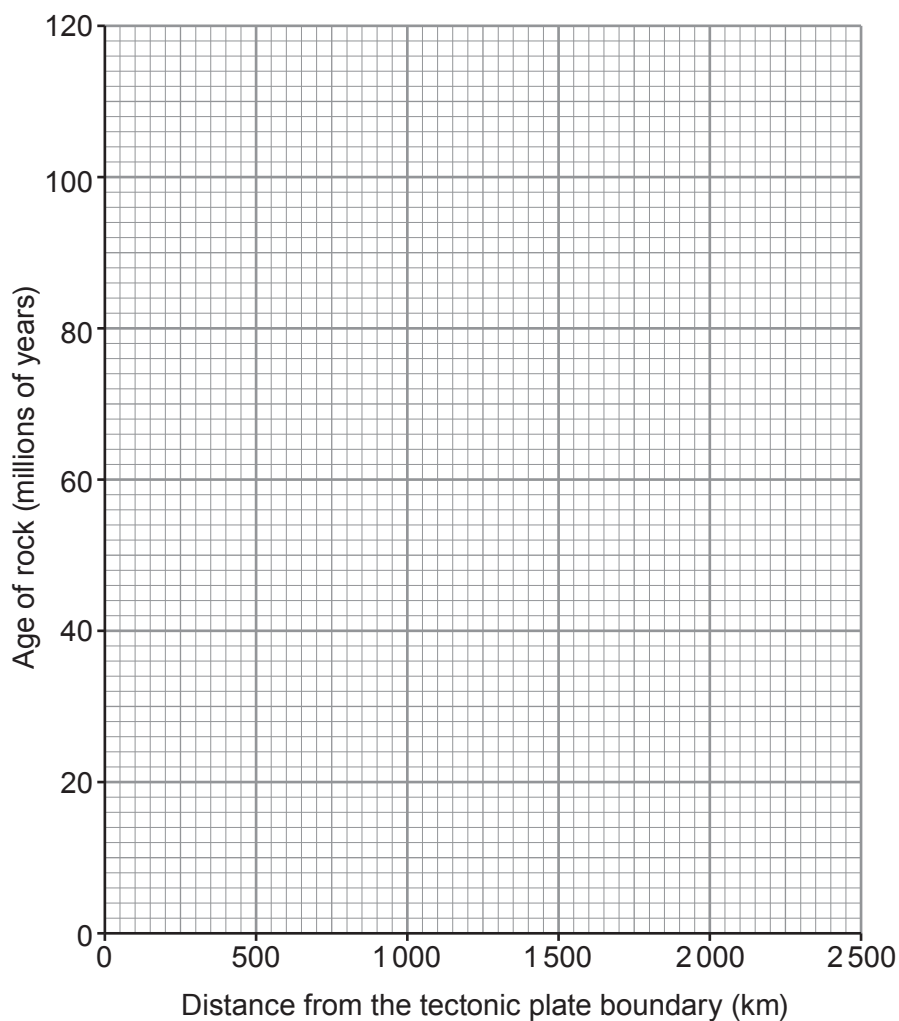
Similar fossils are found on opposite sides of oceans.

☐

- (b) The information in the table below shows the results from a geological survey around tectonic plate boundaries.

Distance from the tectonic plate boundary (km)	Age of rock (millions of years)
0	0
500	24
1000	45
1500	70
2000	90
2500	115

- (i) Use the data in the table above to plot a graph on the grid below and draw a suitable line. [3]



- (ii) Use your graph to estimate the age of rocks that are 750 km from the tectonic plate boundary. [1]

Age = millions of years.

- (iii) Describe the trend shown by the graph. [1]

.....

.....

- (c) Use **only** the words in the box to complete the paragraph below. [3]

mantle	crust	core	convection	conduction	molten
--------	-------	------	------------	------------	--------

The Earth's is broken into large pieces called tectonic plates. These are constantly moving at a distance of a few centimetres each year. Over millions of years the movement has allowed whole continents to shift. This process is called continental drift. The plates move because of currents in the Earth's

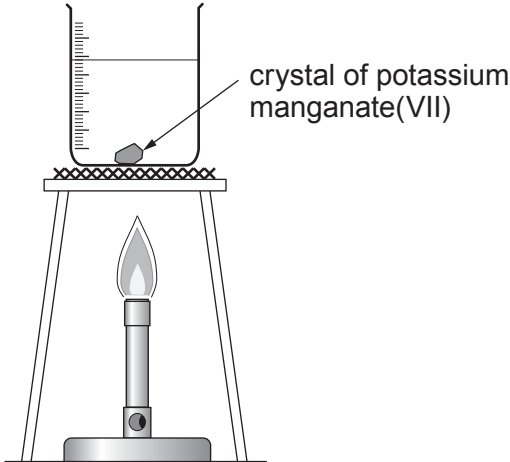
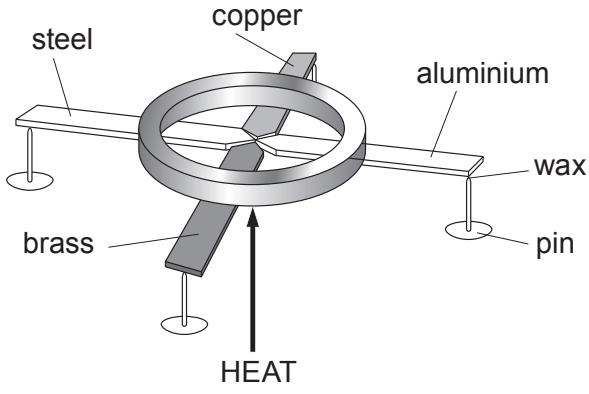
- (d) State **two** natural disasters that can take place at a plate boundary. [2]

1.

2.

2. Heat can be transferred through materials by conduction, convection and radiation. Three groups of students carried out two experiments to show different methods of heat transfer.

They used the following equipment to perform the two experiments.

Experiment 1	Experiment 2
Experiment to show heat transfer through the liquid by  crystal of potassium manganate(VII)	Experiment to show heat transfer through metals by  steel copper aluminium wax pin brass HEAT

- (a) **Complete the diagrams** above by identifying the type of heat transfer shown by each experiment. [2]
- (b) The results of **Experiment 2** are shown in the table below.

Metal	Time taken for the pin to drop off (s)			
	Group 1	Group 2	Group 3	Mean
Steel	73	78	74	
Copper	14	15	13	14
Aluminium	20	19	18	19
Brass	35	37	46	36

- (i) **Complete** the table. [1]
- (ii) **Circle** the anomalous result. [1]

- (iii) Use the results to arrange the metals in order starting with the one that transfers heat the best. [1]

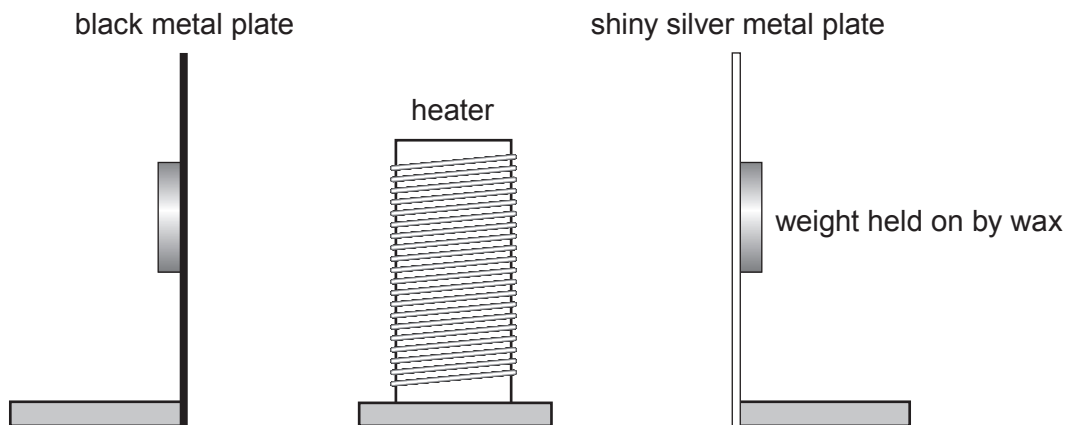
Best metal for transferring heat

.....

brass

Worst metal for transferring heat

- (c) The students carried out another experiment as shown in the diagram below.



Explain why this experiment shows that the plates are heated by radiation and not by convection or conduction. [3]

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.....

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3. The water that is used in the home goes through a number of treatment steps.

- (a) The steps that water undergoes before it passes into the home are given below. They are **not** in the correct order.

Step	Process	Description
A	Distribution	Water is transported through pipes underground
B	Filtration	Water is filtered through layers of sand and gravel removing small insoluble particles
C	Sedimentation	Large insoluble particles settle out by gravity
D	Chlorination	Water is disinfected to kill any remaining bacteria to make it safe to drink
E	Use	Water is used in the home
F	Collection	Water is collected and stored in reservoirs.

Arrange the steps in the **correct** order by completing the boxes below.
The first and last boxes have been completed for you.

[3]



- (b) Pollutants can be found in drinking water making it unsafe to drink. Three students, Andrew, Robert and Malcolm investigated the solubility of a number of different salts at different temperatures.

Describe a method they could follow to find the solubility of a salt at different temperatures. [6 QER]

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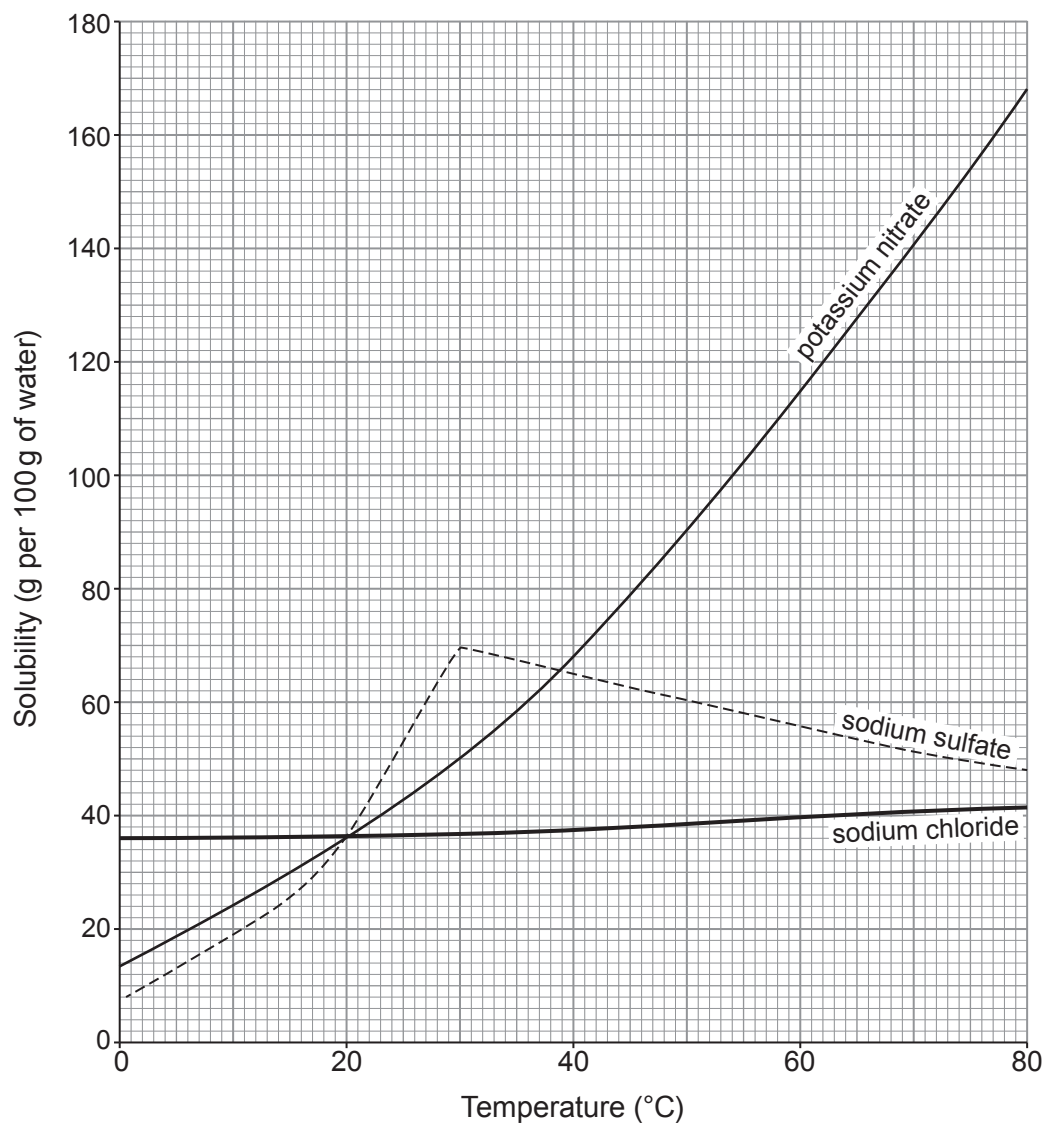
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- (c) The results of an investigation into the solubility of three salts are shown in the graph below.



- (i) State which salt is the most soluble at 30 °C.

[1]

.....

- (ii) Calculate the difference in solubility between potassium nitrate and sodium chloride at 60 °C.

[1]

Answer = g/100g of water

- (iii) Compare the solubilities of potassium nitrate and sodium sulfate between 30 °C and 60 °C. [3]

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- (d) Sam, a school technician, found a salt in an unlabelled container. She followed the same method as the students and her results are shown below.

Temperature (°C)	Solubility of unknown salt (g per 100 g of water)
10	24.0
30	52.0
50	120.0

Sam compared her results with the graph and identified the salt as potassium nitrate. Explain why Sam made this conclusion and whether she was correct to do so. [2]

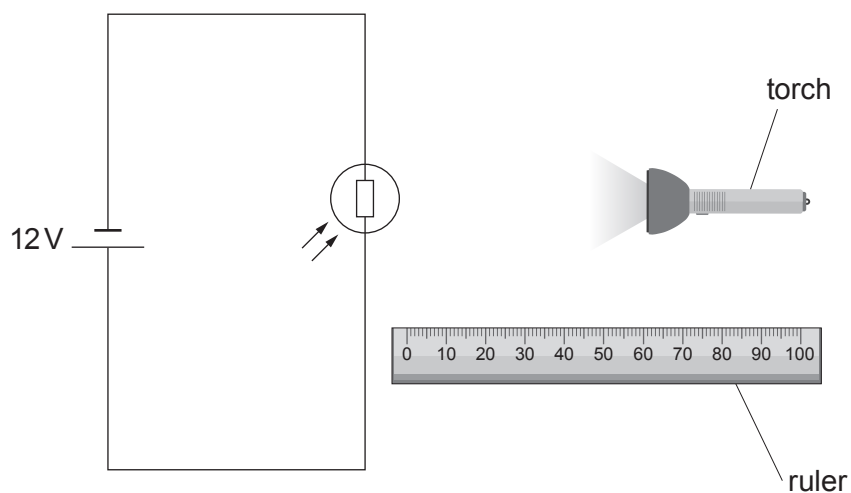
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4. Dafydd, a year 10 student, set up a circuit to investigate the properties of a light dependent resistor (LDR). He changed the distance between a torch and the LDR. He then took measurements of current and voltage to calculate the resistance of the LDR.



- (a) (i) **Complete** the circuit above by adding an ammeter to measure the current through the LDR and a voltmeter to measure the voltage across the LDR. [2]
- (ii) The results of Dafydd's experiment are shown in the table below.

Distance of the torch from the LDR (cm)	Resistance of the LDR (Ω)
2	400
4	800
6	1000
8	1200
12	1300
16	1800
20	2000

Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the **current** passing through the LDR when the light source is 4 cm away. [2]

Current = A

- (iii) Dafydd claims that for every 2cm he moves the torch away from the LDR its resistance doubles. Use Dafydd's results to explain whether his claim is correct. [2]

.....

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- (b) Emyr, another student, also carried out the experiment, but he used a 6 V battery. **Complete** the following sentence by underlining the correct phrase in the brackets. [1]

The resistances of the LDR in the table (**would be greater / stayed the same / would be less**).

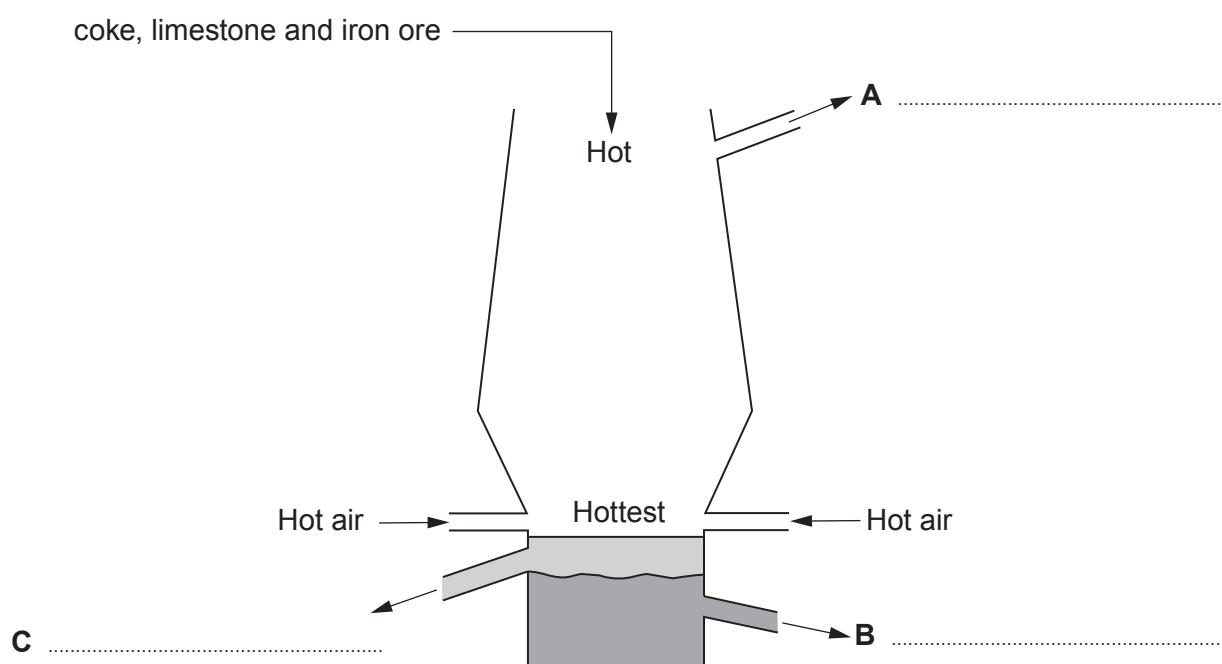
5. Port Talbot steel works in South Wales is one of the largest in the UK. It is capable of producing 5 million tonnes of steel every year.

A number of raw materials are used in a blast furnace to produce iron.

- (a) The diagram below shows the blast furnace and the substances that are added. Hot waste gases, molten iron and slag are the products formed.

Complete the labelling of the diagram.

[2]



- (b) The table below shows some information about the substances used in the blast furnace.

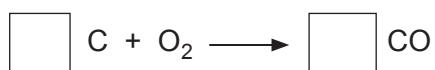
Complete the table.

[4]

Substance	Chemical name	Chemical formula	Type of material
iron ore		Fe_2O_3	compound
coke	carbon		
limestone	calcium carbonate	CaCO_3	

- (c) The blast furnace relies on the processes of reduction and oxidation.

Complete the balanced symbol equations for two of the processes occurring in the blast furnace. [3]



- (d) The table below shows the cost of the raw materials required to produce one tonne of iron in the blast furnace.

Raw material	Mass of raw material (tonnes)	Cost per tonne of raw material (£)	Total cost of raw material (£)
iron ore	1.75	80	
coke	0.25	120	30
limestone	0.25	60	15
air	4.00	3	12
Total cost to produce one tonne of iron			£

Complete the table above to calculate the total cost of materials required to produce one tonne of iron. [2]

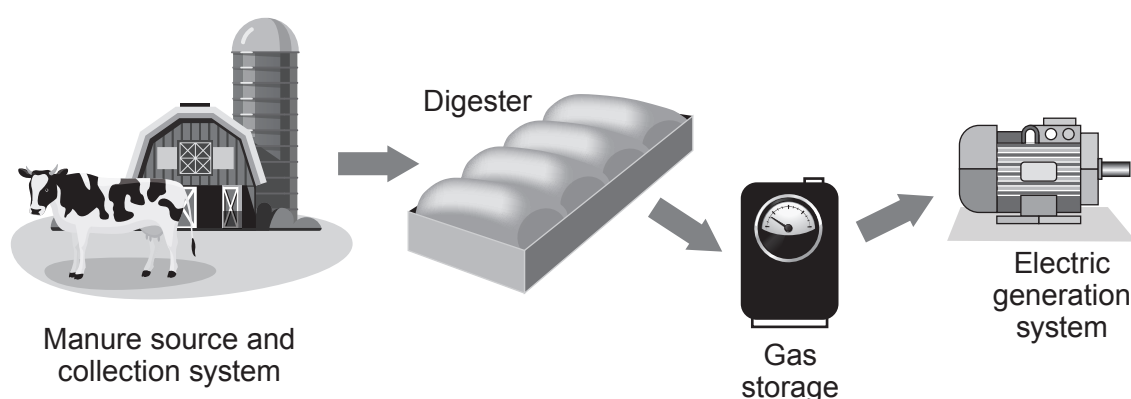
- (e) Aluminium is also extracted from its ore but by a different process. Explain why aluminium cannot be extracted in a blast furnace using coke (carbon). [1]

.....

6. A farmer has installed a biogas generator to save money. The farmer's monthly electricity bill before installation was £3 000. The farmer spent £120 000 to buy and install the biogas generator. After installation the monthly electricity bill was expected to reduce to £600.

The biogas generator works by using the animal waste produced on the farm. This waste is digested by bacteria and the product, methane gas, is used to generate electricity. Methane is a greenhouse gas and produces carbon dioxide and water when it is burned.

The animal waste used in the digester is a renewable energy source.



- (a) The biogas generator can provide some of the farm's electricity.

Use the information above to calculate the expected payback time for the biogas generator. [3]

Payback time = months

- (b) The farmer considered installing wind turbines to generate electricity before investing in the biogas generator.

Give **two** disadvantages of using wind power rather than the biogas generator. [2]

.....

.....

- (c) The biogas generator needs to produce 160 000 kWh of electricity per year to power the farm. A typical cow produces 10 tonnes of waste per year of which the farmer is only able to collect 3 tonnes. For every tonne of waste collected 480 kWh of heat energy is produced, which generates 160 kWh of electricity.

- (i) Use the equation:

$$\% \text{ Efficiency} = \frac{\text{energy usefully transferred}}{\text{total energy supplied}} \times 100$$

to calculate the efficiency of this biogas generator

[2]

% Efficiency =

- (ii) The farmer owns 120 cows. He thinks that he will be able to collect enough waste to power his farm for a year. [4]

Explain whether you agree with the farmer. Show your workings as part of your answer.

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7. In a diffusion experiment, gelatin was mixed with a pink pH indicator and allowed to set. The gelatin was then cut into different sized cubes. The cubes were placed in a beaker containing dilute acid. As the acid diffused through the gelatin the cubes became colourless. The time taken to change from pink to colourless was recorded.

The surface area and volume of the cubes were calculated using the following equations:

$$\text{surface area of cube} = \text{length} \times \text{width} \times \text{number of sides}$$

$$\text{volume of cube} = \text{length} \times \text{width} \times \text{height}$$

- (a) The results of the experiment are shown below.

Length of side of gelatin cube (cm)	Time taken to become colourless (s)	Surface area of cube (cm ²)	Volume of cube (cm ³)	Surface area:volume ratio
1	20	6	1	6:1
5	41	150		6:5
7	104		343	6:7
10	610	600	1000	

- (i) **Complete** the table. [3]
- (ii) Sarah suggested that diffusion of the acid into the gelatin is quicker in the smaller cubes because the time taken for them to become colourless is less. Explain whether you agree with Sarah. [2]

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- (b) Red blood cells transport oxygen around the body to allow respiration to occur in the tissues. Red blood cells have a volume of 90 units and a surface area of 136 units. Explain how the surface area:volume ratio of red blood cells make them suitable for their function. [3]

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END OF PAPER

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1										4 He Helium 2							
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89	Key														

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445U10-1



MONDAY, 20 JUNE 2022 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

FOUNDATION TIER

1 hour 30 minutes

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **3** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A Periodic Table is printed on page 20.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	10	
3.	6	
4.	12	
5.	8	
6.	12	
7.	19	
Total	75	

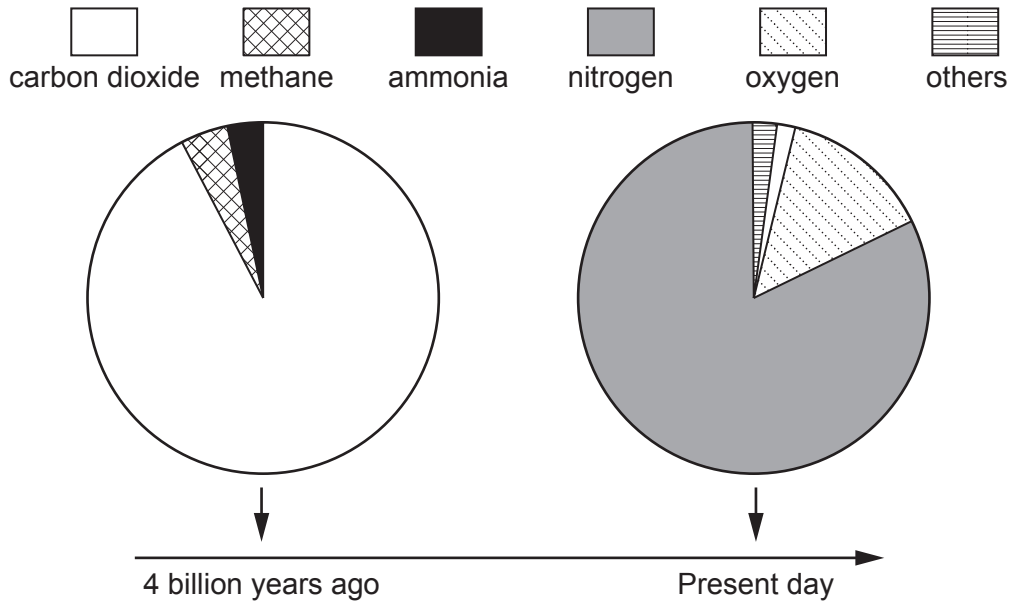
3445U101
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JUN23445U10101

Answer **all** questions.

1. The pie charts show the composition of the air on the Earth 4 billion years ago and the present day.



- (a) (i) Give **two** reasons why the amount of carbon dioxide has changed so much over this time period. [2]
1.
2.
- (ii) State **three** other changes shown by the pie charts. [3]
1.
2.
3.



(b) The table below shows some of the other gases in the air during the present day.

Gas	Chemical formula	Approximate percentage composition (%)
argon	Ar	1.0
hydrogen	H ₂	0.00005
methane	CH ₄	0.0002
neon	Ne	0.002
nitrous oxide	N ₂ O	0.00003
ozone	O ₃	0.000004

Tick (✓) **three** correct statements about the gases in the air.

[3]

Nitrous oxide contains carbon atoms

☐

Ozone only contains oxygen atoms

☐

Hydrogen is an element that occurs as a molecule

☐

There is a higher percentage of nitrous oxide than methane

☐

The nitrous oxide molecule contains two oxygen atoms

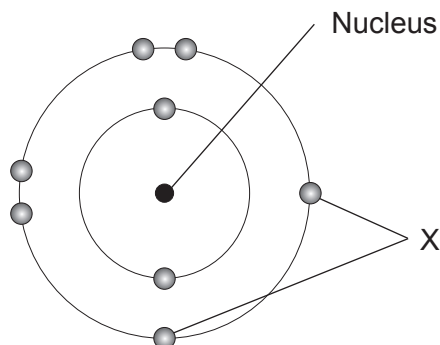
☐

Methane is a hydrocarbon

☐


2. The Periodic Table is an arrangement of the chemical elements according to their atomic number, electron configuration and chemical properties.

An oxygen atom is shown below.



- (a) Complete the following sentences using **only** the words in the box. The words may be used once, more than once, or not at all.

protons	ions	electrons	neutrons
---------	------	-----------	----------

- (i) The nucleus of the oxygen atom contains and

.....

[2]

- (ii) The particles labelled X are called

[1]

- (iii) In a neutral atom the number of protons is equal to the number of

.....

[1]

- (b) Oxygen is found in Group 6 of the Periodic Table. Use the Periodic Table on page 20 to answer the following questions.

- (i) Give the electron configuration of oxygen.

[1]

.....

- (ii) State the name of **one** element in the same **group** as oxygen.

[1]

.....

- (iii) State the name of **one** element in the same **period** as oxygen.

[1]

.....



- (c) Water is a compound that contains one oxygen atom and two hydrogen atoms joined together.

- (i) Use the Periodic Table on page 20 to complete the following table. [2]

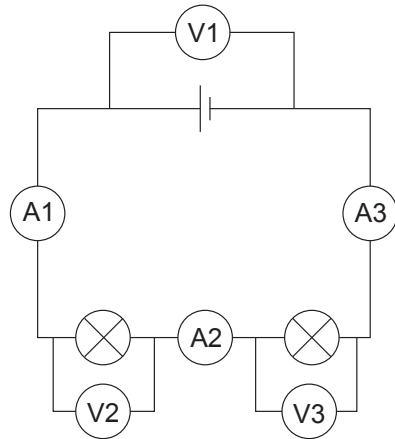
Element	Relative atomic mass
hydrogen	
oxygen	

- (ii) Use the information above to calculate the relative formula mass of H_2O . [1]

relative formula mass =



series circuit



- the readings on the ammeters
- the readings on the voltmeters.

[6 QER]



4. The UK government aims to remove the sale of petrol and diesel cars by 2030. It is expected that by 2027 more electric cars will be sold than any other type.

- (a) The data in **Table 1** shows how the power generated in a typical petrol engine is converted.

Table 1. Efficiency of a petrol driven car

	Description of energy conversion	% Power
engine	heat loss	70
movement	noise	1
	friction	6
	power to the wheels	
	total power	100

- (i) Use the information in **Table 1** to calculate the % power to the wheels. [1]

% power to the wheels =

- (ii) A petrol car generates 66 kW of power.

Use your answer to (a)(i) to calculate the actual power delivered to the wheels. [2]

actual power to the wheels = kW



- (iii) State **two** ways in which engineers have improved the design of cars to improve their efficiency. [2]

1.

.....

2.

.....

- (b) One of the biggest problems with electric vehicles is the distance travelled per charge.

Table 2 gives information about three types of battery chargers in use.

Table 2

Type of charger	Charger 1	Charger 2	Charger 3
operating voltage (V)	V_1	240	480
operating current (A)	20	80	I_3
power delivery (W)	2400	19 200	144 000
distance travelled per hour of charging (km)	3	83	545

- (i) Use the equation

$$\text{voltage} = \frac{\text{power}}{\text{current}}$$

to calculate the operating voltage, V_1 , of charger 1.

[2]

$$V_1 = \dots\dots\dots \text{ V}$$



(ii) Use the equation

$$\text{current} = \frac{\text{power}}{\text{voltage}}$$

to calculate the current, I_3 , of charger 3.

[2]

$I_3 = \dots\dots\dots$ A

(iii) Electric car companies claim that it is cheaper to run their electric cars than petrol ones.

Use the following information to explain if this claim is correct.

[3]

- A typical petrol car needs 50 litres of fuel to travel 545 km. The cost of one litre of petrol is £1.30.
- An electric car requires 144 kWh of electricity to travel 545 km. The cost of one unit of electricity is £0.30.
- cost of electricity (£) = cost per unit (£) $\times$ number of units (kWh)

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09

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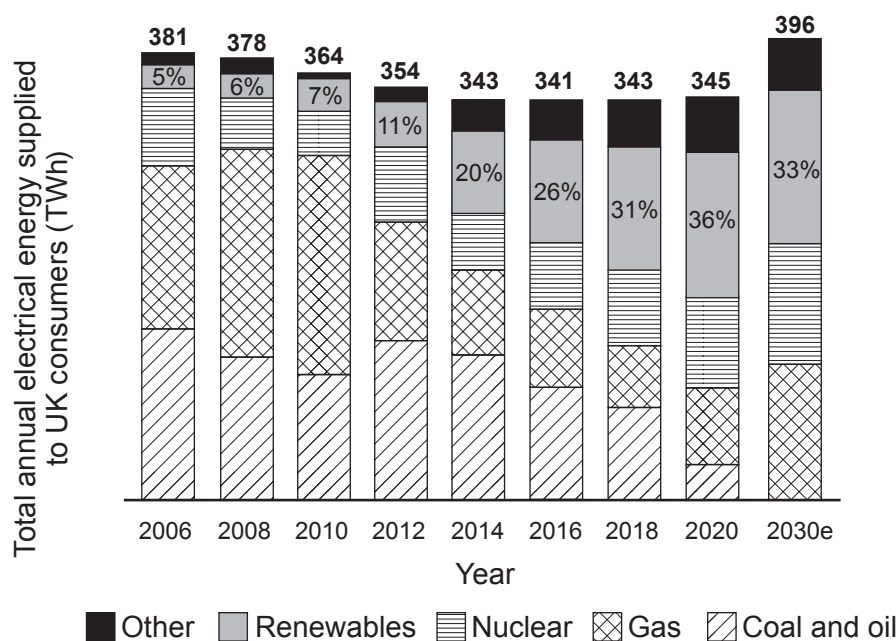
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5. The graph below shows the total annual electrical energy supplied to UK consumers and the source of generation.

The total energy is given in terawatt hours (TWh).

The figures for 2030 are estimates.



- (a) Use the graph to answer the following questions.

- (i) Describe how the use of gas has changed between 2006 and 2018. [2]

.....

.....

.....

- (ii) Calculate how much extra electrical energy will need to be supplied to the UK in 2030 compared to 2020. [1]

extra electrical energy = TWh



- (iii) Calculate how much electrical energy is expected to be generated from renewable sources in 2030. [2]

renewable electrical energy generated = TWh

- (b) One of the largest coal-fired power stations in the UK can generate 950 MW. When it was built the engineers thought it could supply at least 400 000 homes.

The mean power delivered to a house is 1840 W.

Explain whether you agree with the engineers. [3]

(1 MW = 1 000 000 W)

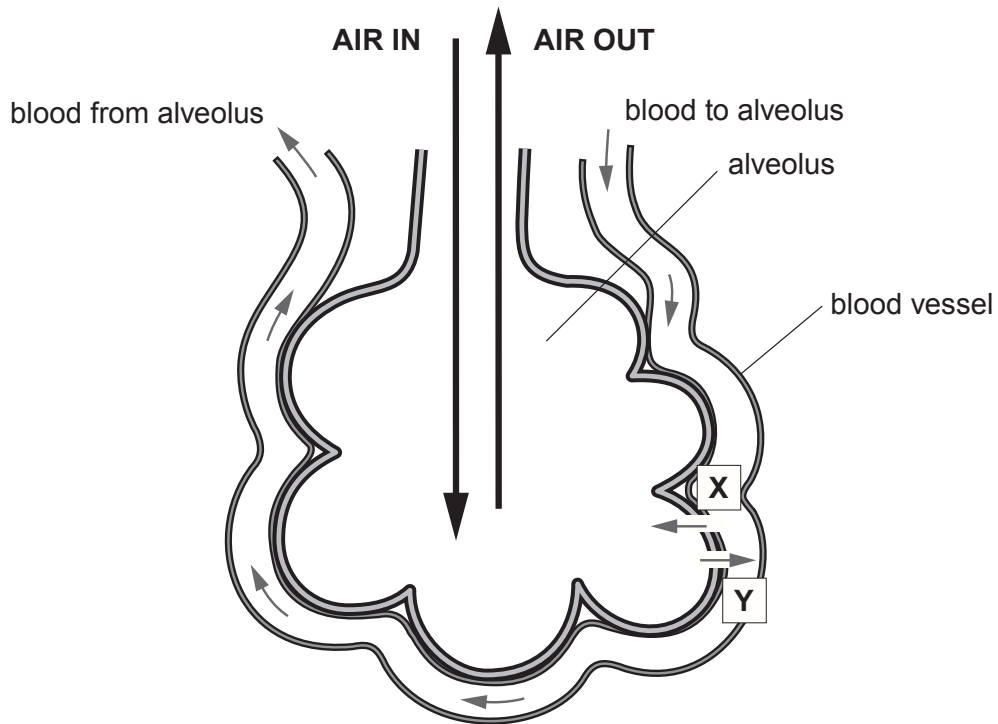
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6. The human body needs a respiratory system for gas exchange.

(a) The diagram below shows an alveolus.



- (i) Gases **X** and **Y** diffuse into and out of the alveolus. Name the gases **X** and **Y** in the diagram.

[1]

X

Y

- (ii) Name the type of blood vessel that surrounds the alveolus.

[1]

.....



- (b) Jade is an Olympic athlete. She trains on a regular basis. Training improves her respiratory system.

The table shows information about Jade's breathing at rest and at exercise during her latest test.

	Breathing rate (breaths per minute)	Depth of breathing (cm ³ per breath)	Total volume of air breathed in (cm ³ per minute)
at rest	12	500	
during exercise	40		40 000

- (i) Complete the table. [2]

- (ii) Describe the changes that exercise has on Jade's breathing. [2]

.....

.....

.....

- (iii) Jade's coach wants her lungs to be able to absorb at least 2000 cm³ of oxygen per minute as this will help with aerobic respiration in her muscles.

Inhaled air contains 21% oxygen and exhaled air contains 16% oxygen.

Jade thinks she has achieved this in her latest test.

- Use the information given to determine whether you agree with Jade. [3]

.....

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.....



(c) The oxygen absorbed is used in the cells to release energy by aerobic respiration.

- (i) During intense exercise some of Jade's muscles use the process of **anaerobic** respiration. State what is meant by the term 'anaerobic respiration'. [1]

.....

.....

- (ii) Give **one advantage** and **one disadvantage** of anaerobic respiration occurring in Jade's muscle cells. [2]

Advantage:

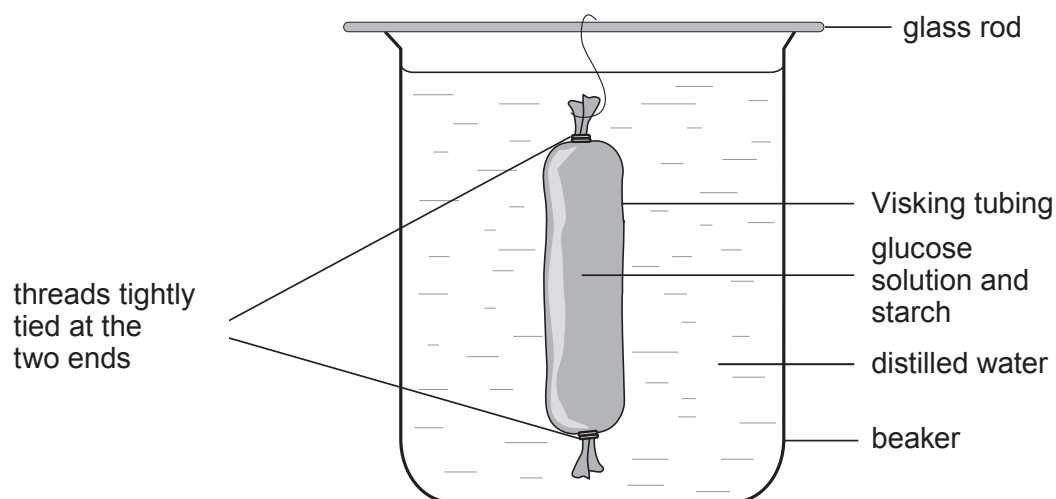
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Disadvantage:

.....



7. A student set up the following experiment to show how diffusion of digested foods occurs in the small intestine. The Visking tubing contained starch and a solution of glucose with a concentration of 100 g/dm^3 . This was placed in a beaker containing distilled water at 20°C and left for 2 hours.



- (a) (i) Explain what is expected to happen to the starch and the glucose in the Visking tubing during the experiment. [4]

.....

.....

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.....

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.....

- (ii) Describe **two** chemical tests the student should carry out on the contents of the beaker to confirm your answer to (a)(i) and state the expected results. [4]

1.

.....

.....

2.

.....

.....



(b) Explain how each of the following changes would affect the diffusion process in this experiment.

(i) Carbohydrase is added to the contents of the Visking tubing. [2]

.....

.....

.....

(ii) The investigation is carried out at 35°C. [2]

.....

.....

.....

(iii) The experiment is carried out with a solution of glucose of 100 g/dm³ concentration replacing the distilled water in the beaker. [2]

.....

.....

.....

(c) State **one** of the limitations of using Visking tubing as a model intestine. [1]

.....

.....

(d) Explain **two** adaptations of the small intestine for the absorption of digested food. [4]

1.

.....

.....

2.

.....

.....

END OF PAPER

19





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atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445U10-1



S23-3445U10-1

FRIDAY, 16 JUNE 2023 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

FOUNDATION TIER

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	14	
3.	9	
4.	12	
5.	11	
6.	8	
7.	11	
Total	75	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **3(b)** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A Periodic Table is printed on page 20.



JUN233445U10101

Answer **all** questions.

1. Ffion is investigating a neutralisation reaction between sulfuric acid and sodium hydroxide.

- (a) Use the correct words from the box to complete the **word** equation for this reaction below. [1]

sodium chloride	sodium nitrate	sodium sulfate
-----------------	----------------	----------------

sulfuric acid + sodium hydroxide $\longrightarrow$ + water

- (b) In the experiment she monitors the change of pH using both a pH meter and an indicator.

The chart below shows the colour of phenolphthalein indicator at different pH.

pH scale	1	2	3	4	5	6	7	8	9	10	11	12	13	14
phenolphthalein	colourless								pale pink	pink				

- (i) The table below shows Ffion's results. Complete the table. [2]

Volume of sodium hydroxide added (cm ³)	pH meter reading	Phenolphthalein indicator colour
0	1.0	colourless
5	1.0	colourless
10	1.1	colourless
15	1.2	colourless
20	2.0	
25	9.0	pale pink
30	13.0	
35	14.0	pink
40	14.0	pink



- (ii) Use the information in the table to tick (✓) the boxes next to the **three** correct statements. [3]

Before any sodium hydroxide is added to the acid the phenolphthalein turns green.

☐

The pH is lowest at the start of the experiment.

☐

When 10 cm³ of sodium hydroxide has been added the solution is neutral.

☐

As sodium hydroxide is added the pH rises.

☐

The pH is highest at the end of the experiment.

☐

When 40 cm³ of sodium hydroxide has been added the solution is acidic.

☐

- (c) Sodium hydroxide (NaOH) contains the three elements sodium, hydrogen and oxygen.

- (i) Select the correct numbers from the box to complete the table below. [3]

1	2	8	11	12	16
---	---	---	----	----	----

Element	Number of protons	Number of neutrons	Number of electrons	Mass number
sodium	11	12		23
oxygen	8		8	16
hydrogen	1	0	1	

- (ii) Sodium metal is found in Group 1 of the Periodic Table. Use the Periodic Table on page 20 to name **one** other metal found in this group. [1]

.....



2. The UK has been generating more energy using wind power over recent years.



- (a) Circle **one** renewable energy source below.

[1]

coal

gas

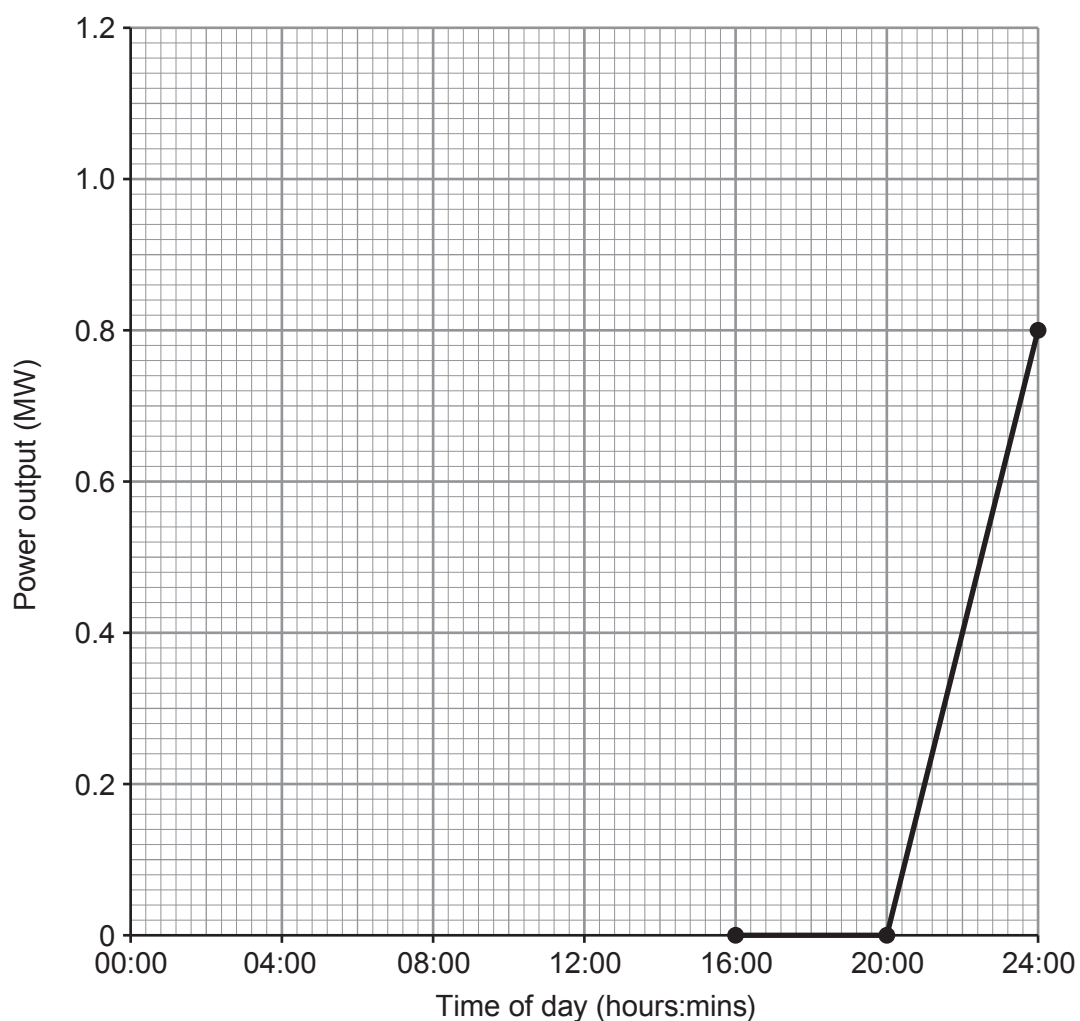
solar

- (b) The information below shows the power output of a wind turbine in one day.

Time of day (hours:mins)	Power output of wind turbine (MW)
00:00	0.6
04:00	1.0
08:00	1.2
12:00	0.8
16:00	0.0
20:00	0.0
24:00	0.8



- (i) Use the data from the table to complete the graph on the grid below and join the points with a ruler. The last three points have been plotted for you. [3]



- (ii) State why the power generated by this wind turbine is not constant during this day. [1]

.....

.....



(iii) Use the information to answer the following questions.

I. State the maximum power produced by the wind turbine during this day. [1]

maximum power = MW

II. State the time of day that the wind speed was greatest. [1]

time =

III. Determine between which hours the wind turbine did not generate any electricity. [1]

from: to:

(iv) David suggested that the wind turbine generated a power greater than 0.6 MW for 16 hours.

Add a horizontal line to your graph and use it to explain whether David was correct. [4]

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.....

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.....

(c) Last year this wind turbine had a mean power output of 0.6 MW. A typical nuclear power station has a continuous power output of 3600 MW.

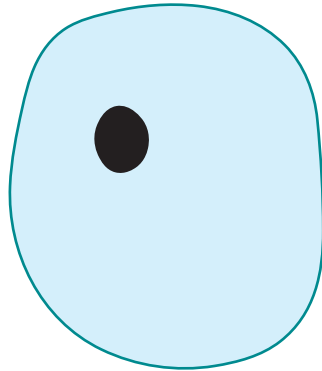
Calculate how many wind turbines would be needed to generate the same power output as a nuclear power station. [2]

number of wind turbines =

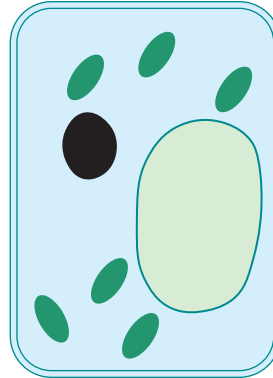


3. Clara is studying the structure and function of cells. The images below show animal and plant cells.

Animal cell



Plant cell



- (a) Complete the table below to show which parts are present in animal cells.

[3]

Part	Plant cells	Animal cells (✓ or ×)
cell wall	✓	×
cytoplasm	✓	✓
chloroplast	✓	×
nucleus	✓	
cell membrane	✓	
vacuole	✓	



(b) Describe the functions of the parts of a **plant cell** listed in the table.

[6 QER]

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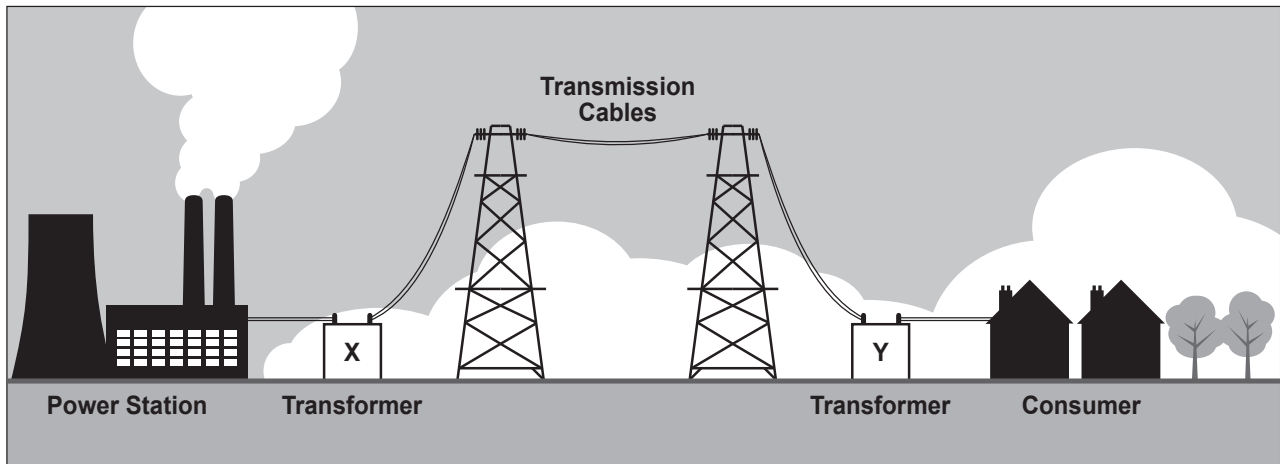
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4. The diagram below shows part of the National Grid.



- (a) Complete the sentences about the National Grid by selecting the correct words from the box. Each word can be used once, more than once or not at all. [4]

faster reliable increases generates safely decreases

The National Grid makes sure that the electricity supply to consumers is

.....

Transformer X the voltage and
the current.

Transformer Y changes the voltage so that consumers can use it
..... in their homes.



- (b) Five electrical supply companies (**A**, **B**, **C**, **D** and **E**) each produce the same amount of electrical power.

The table below shows the percentage of electrical power output from each energy source by each company.

Energy source	Percentage of electrical power output from each energy source (%)				
	Company A	Company B	Company C	Company D	Company E
coal	4.8	5.8	7.1	10.8	10.1
natural gas	63.0	55.5	9.6	63.0	46.0
nuclear	5.2	20.6	71.2	23.3	13.9
renewable	27.0	18.1	12.1	2.9	30.0

Use the table to answer the following questions.

- (i) State which energy company produces the most electrical power from coal. [1]

.....

- (ii) State which company produces the most nuclear waste. [1]

.....

- (iii) Calculate the **mean** percentage of electrical power output generated from **coal** by the five companies (**A**, **B**, **C**, **D** and **E**). [2]

mean percentage =

- (c) The carbon footprint of an energy supply customer can be an important factor when choosing a supply company.

- (i) State what is meant by the term 'carbon footprint'. [2]

.....

- (ii) A customer decides against using Company **A** since they believe that they have the highest carbon footprint from generating electricity.

Use information from the table to explain whether the customer is correct. [2]

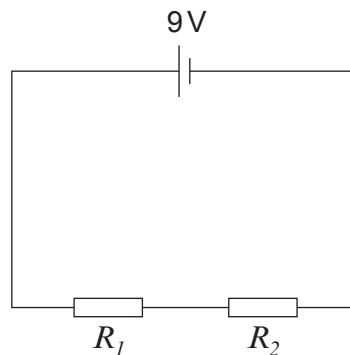
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5. Jamie is investigating an electrical circuit which has two **identical** resistors, R_1 and R_2 . The circuit that he built is shown below.



- (a) Complete the sentences by underlining the correct word(s) in the brackets.

- (i) In this series circuit the current in R_1 is (**the same as / higher than / lower than**) the current in R_2 . [1]
- (ii) In this series circuit the voltage across R_1 is (**the same as / higher than / lower than**) the voltage across R_2 . [1]

- (b) The voltage across R_1 is 4.5 V.

- (i) Each resistor was found to have a resistance of $2.25\ \Omega$.

I. Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in R_1 .

[2]

current = A

- II. Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power developed in resistor R_1 .

[2]

power developed in R_1 = W

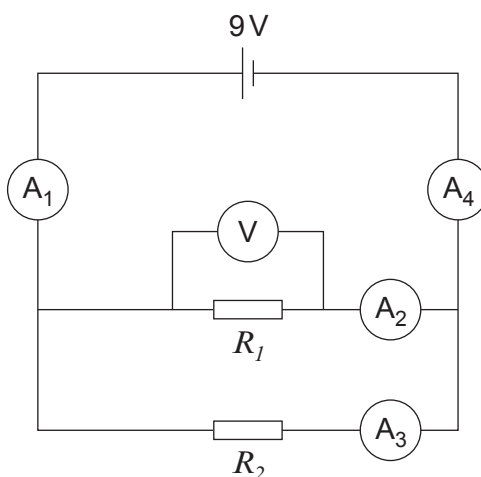


(ii) State the voltage across R_2 .

[1]

voltage across $R_2 = \dots\dots\dots$ V

(c) Jamie then built a parallel circuit using the same 2.25Ω resistors.



(i) State which ammeter (A_1 , A_2 , A_3 or A_4) measures the current through R_1 .

[1]

$\dots\dots\dots$

(ii) State the voltmeter reading across R_1 .

[1]

voltmeter reading = $\dots\dots\dots$ V

(iii) Ammeter A_2 reads 4 amps.

I. State the reading on ammeter A_3 .

[1]

ammeter reading = $\dots\dots\dots$ A

II. Calculate the reading shown on ammeter A_1 .

[1]

ammeter reading = $\dots\dots\dots$ A



6. Sioned reacted sulfuric acid with zinc oxide (ZnO).
She followed the method below to form zinc sulfate.

1. Pour 50 cm³ of 1.0 mol/dm³ sulfuric acid into a conical flask.
2. Heat the sulfuric acid to 50°C.
3. Add 5 g of zinc oxide to the warmed acid and stir.
4. Repeat step 3 until the zinc oxide is in excess.

- (a) The main hazard in this experiment is that 1.0 mol/dm³ sulfuric acid is an irritant.

- (i) Describe **one** risk associated with using sulfuric acid.

[1]

.....

.....

- (ii) State a suitable control measure Sioned should carry out to reduce this risk.

[1]

.....

.....

- (b) Circle the correct formula in each bracket to complete the symbol equation for this reaction.

[2]



- (c) State **two** additional steps that Sioned needs to carry out to obtain pure zinc sulfate crystals.

[2]

1.

.....

2.

.....

- (d) Robert has calculated the relative formula mass of zinc oxide to be 97.
Use the Periodic Table on page 20 to explain if he is correct.

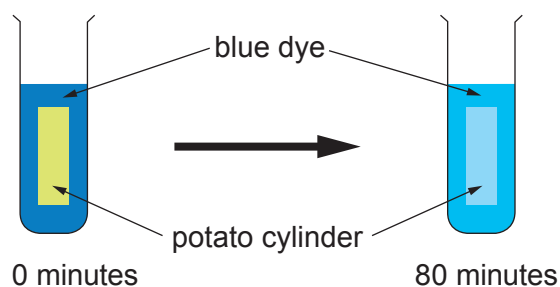
[2]

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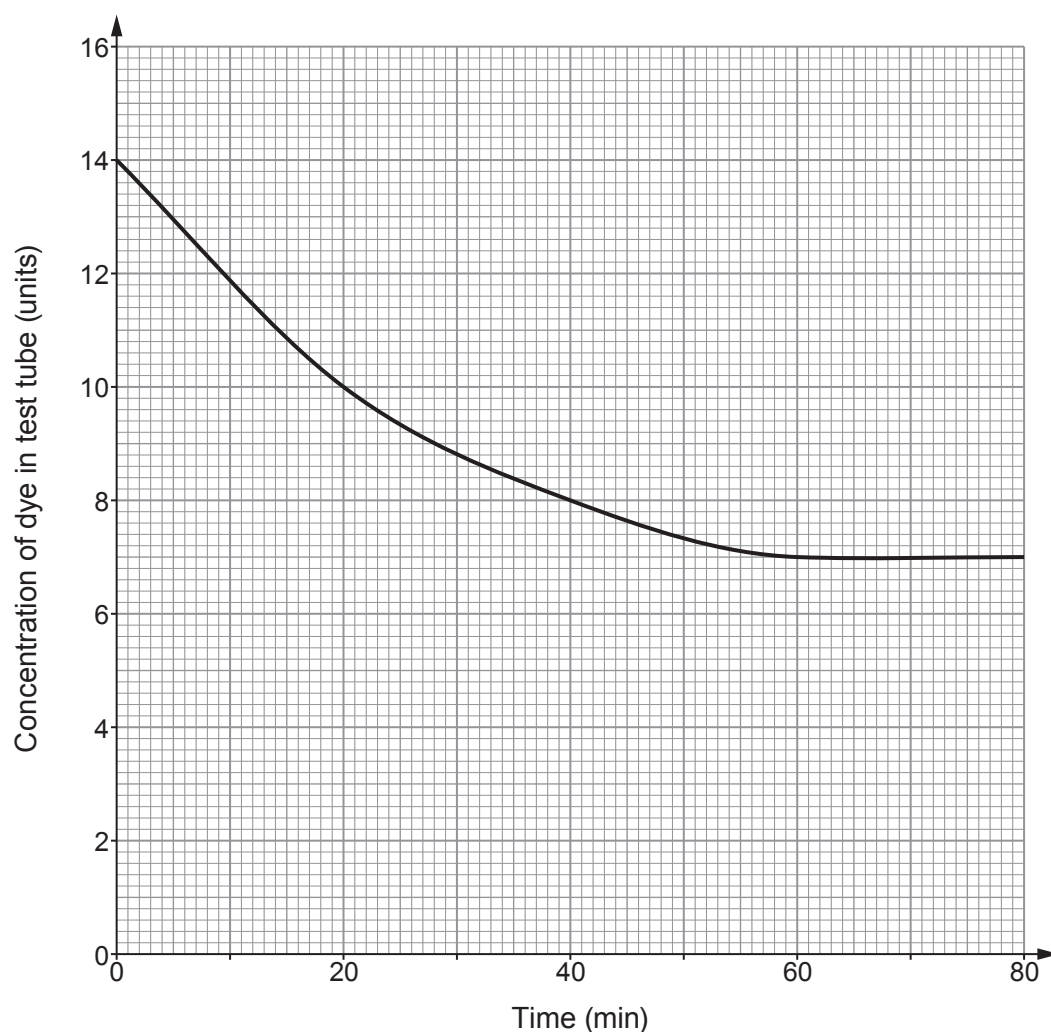


7. Several groups of students investigated the rate of diffusion of blue dye into potato by the method below.

1. Cut a cylinder of potato.
2. Place the cylinder in a test tube containing a solution of blue dye.
3. Put the test tube in a water bath at 20°C.
4. Measure the concentration of the dye after 20 minutes.
5. Continue to measure the concentration every 20 minutes up to 80 minutes.



The concentration of the dye left in the test tube after each 20-minute interval is shown on the graph below.



- (a) (i) State the dependent variable in this experiment. [1]

.....

- (ii) State **two** variables that should be controlled. [2]

1.

2.

- (b) (i) State how the concentration of dye left in the test tube changed during the first 20 minutes. [1]

.....

- (ii) Explain why this change occurred. [2]

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- (c) One student suggests that the rate of diffusion is constant over the whole 80 minutes. Explain whether you agree. [3]

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- (d) The students repeated the experiment in a water bath at 30°C. **Add another line** to the graph to show the expected results. [2]

END OF PAPER





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atomic number

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